



MY MATHEMATICS

Grade 5

Government of Nepal
Ministry of Education
Curriculum Development Centre



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Foreword

Continuity on the process of development and updating of curriculum and textbooks has been given so as to make school level education objective oriented, practical, contemporary and employment oriented. Fostering the feelings of unity towards the nation & nationality, developing attributes like morality, discipline, self-reliance, promoting fundamental language related and mathematical skills through the basic knowledge of science, environment and health-related issues. Education should instill life skills, create interest in art and beauty promote tolerance among different ethnicities, gender, religion, language, culture. Likewise it should make learners responsible citizen who can safeguard social values and norms. To fulfill this need this textbook has been revised and developed based on the curriculum 2065 which was updated according to the suggestions of various level committees, teachers, guardians and the final decisions of seminars and interactions including other stakeholders of education.

Previously, based on the old curriculum My Mathematics class 5 was written by Sambhu Narayan Vaidya & Hari Narayan Upadhyaya. Re-writing & revision work has been done on it, according to the suggestions of teachers, experts, guardians students & other stakeholders matching it up with the latest styles of national & international textbooks. We have tried to make this text book pictorial, activity oriented and child centered keeping in view with the all round development of grade five students. Based on the curriculum 2065, writing and revision work of this book was done by a panel comprised of Dinesh Kumar Shrestha, Chitra Prasad Devkota, Barun Prasad Vaidya, Hari Narayan Upadhyaya, Dilliswor Pradhan, Dandapani Sharma, Ram Chandra Poudel and Shyam Singha Dhami. Haribol Khanal, Dr. Siddhi Prasad Koirala, Dr. Siva Ram Neupane, Mukunda Raj Sharma, Nirmala Gautam were also involved in this work. Content editing was done by Dinesh Kumar Shrestha, Dandapani Sharma & Shyam Singha Dhami & language editing was done by Bishnu Prasad Adhikari and Lok Prasad Pandit. Type setting & layout design was done by Jayaram Kuikel and Ujwal Kundan Jyapu and Himalaya Gautam did the work of Illustration. Curriculum Development Centre remains grateful to all those who have contributed to the development & revision of this book. This English Version editon has been translated by Krishna Raj Hamal.

Textbook is an important tool of teaching learning process. To achieve the learning outcomes of the curriculum, experienced teachers & inquisitive students will use various resources and materials in teaching learning process. Effort has been put to make this book activity oriented and more interesting, however, there might be some rooms for corrections in its language, presentation style and illustrations. Teachers, students, guardians, intellectuals and general readers of the book can contribute a lot to eliminate these weakness by sending their suggetions. Curriculum Development Centre cordially invites for their constructive suggestions.

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Lesson 2 Concept of Numbers

2.1 Counting of numbers and place value up to hundred million.

There are ten numbers in Hindu-Arabic. They are 0,1,2,3,4,5,6,7,8 and 9.

Let's make a figure using these numbers 36582292. Let's put them in the following place value table. Let's see what is the place value of 5.

Crore	Ten Lakh	Lakh	Ten Thousand	Thousand	Hundred	Ten	One
3	6	5	8	2	2	9	2

Number 5 is at the place value of lakh. So the place value of 5 is 5 lakh = 5,00,000. In the above table number 3 is at the place of crore. Therefore the place value of 3 is 3 crore = 3, 00, 00,000.

Thus the figure is read as three crore, sixty five lakh, eight two thousand, two hundred and ninety two.

In the following table counting figures up to ten crore and the way they are read or their numerical names are given. Look at the table carefully and discuss what you found out:

Made of how many numbers?	Figures	Numerical names
The smallest figure made of one number	1	1(one)
The smallest figure made of two numbers	10	10(ten)
The smallest figure made of three numbers	100	100(Hundred)
The smallest figure made of four numbers	1000	1000(Thousand)
The smallest figure made of five numbers	10000	10000(Ten Thousand)
The smallest figure made of six numbers	100000	100000(Lakh)
The smallest figure made of seven numbers	1000000	1000000(Ten Lakh)
The smallest figure made of eight numbers	10000000	10000000(Crore)
The smallest figure made of nine numbers	100000000	100000000(Ten crore)

Method of writing figures into words:

Figure 106739862 is written in the place value table in the following way:

Figure	Place Value								
	Crore		Lakh		Thousand		Units		
	Ten Crore	Crore	Ten Lakh	Lakh	Ten Thousand	Thousand	Hundred	Ten	One
106739862	1	0	6	7	3	9	8	6	2

We use comma (,) to show the place value so as to make reading a figure easier. So we write figure 106739862 by using comma in this way: 10,67,39,862. While reading it we say: Ten crore, sixty seven Lakh, thirty nine thousand, eight Hundred and sixty two.

Example 1:

Write in words: 19,56,02,183.

19,56,02,183 = Nineteen crore, fifty six lakh, two thousand, one hundred and eighty three.

Method of writing word figures into numbers:

While writing ten crore, sixteen lakh, eight thousand, nine hundred and thirty six into figures we write:

Ten crore, sixteen lakj, eight thousand, nine hundred and thirty six = 10, 16, 08, 936.

Likewise if we have to write ten crore, eighty two lakh, fifteen thousand and fifteen we write: 10, 82, 15, 015.

Example 2:

Write in figures: Eighteen crore, eleven lakh, four thousand and sixty.

Eighteen crore, eleven lakh four thousand and sixty = 18,11,04,060.

Exercise 2.1:

1. Write the place value of 2 in each of the figures below:

- a. 13524 b. 1235497
- c. 2305343 d. 4263578

2. How many lakhs are there in each of the figures below?

- a. 1234567 b. 50031247 c. 67853479

3. How many thousands are there in a crore?

4. Place comma (,) in each of the figures below:

- a. 1350739 b. 2467893 c. 275403016
- d. 75003052 e. 105300274 f. 592070593

5. Write each figure in question 4 into words.

6. Write in figures/ numbers:

- a. One crore, eighty two lakh and fifteen.
- b. Seventeen crore, seventy five lakh, three thousand, six hundred and five.
- c. Twenty two crore, sixty five thousand, seven hundred and seventy.
- d. Thirty six crore, fifty lakh, six hundred and ninety.
- e. Ninety nine crore, five.

2.2 Writing numbers into international system:

International system is different from the Nepali System in placing comma in a figure e.g. 57363542. We keep on placing comma after every three number in international system from the right-hand side. According to this the above figure is written in the following way: 57,363,542.

Placing it in place value table:

Number	Million			Thousand			Unit		
	Hundred	Ten	One	Hundred	Ten	One	Hundred	Ten	One
57363542	-	5	7	3	6	3	5	4	2

We read it as 57 million, 363 thousand and 542.

Or fifty seven million, three hundred sixty three thousand, five hundred and forty two.

Example 1:

Write in words: 123, 430, 316

123, 430, 316 = 123 million, 430 thousand and 316

= one hundred twenty three million, four hundred thirty thousand, three hundred and sixteen.

Writing word figures into numbers in international system

sixty million, three hundred forty two thousand and eighty two.

= 60 million, 342 thousand and 082 = 60, 342, 082.

Example 2:

Write in figures:

One hundred seven million, two hundred ninety three thousand, six hundred and fifty.

= 107, 293, 650.

= 107,293,650

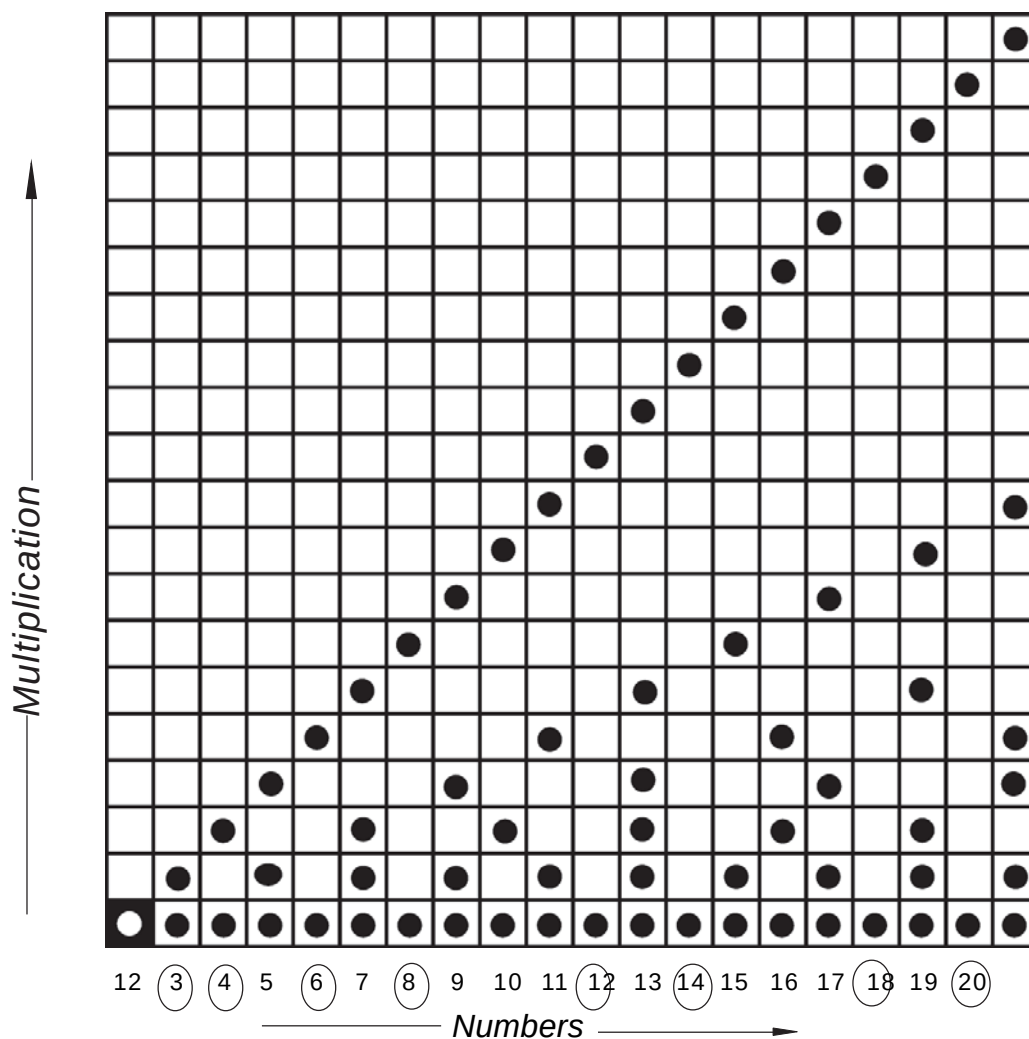
Exercise 2.2

1. Put comma (,) in each of these figures according to the international system:
 - a. 35768229
 - b. 38962352
 - c. 158632932
 - d. 628293563
2. Place it in the place value table and write in words according International System:
 - a. 32, 567, 832
 - b. 178, 625, 123
 - c. 595, 207, 257
 - d. 185, 090, 159
3. Write in figures:
 - a. Fifty seven million, nine hundred twenty-six thousand, one hundred and thirty-three.
 - b. One hundred twenty-four million, sixty-six thousand and three hundred four.
 - c. Two hundred fifty-seven million, four hundred twenty thousand and seventy-four.
 - d. Seven hundred thirty-four million, thirty seven thousand, two hundred and sixty-three.

Tip to the teacher: Ask students to make their own questions like in the above exercise and hold quiz contest in groups.

2.3 Prime and composite numbers

Given below are the prime and composite numbers from 1-20. Dot used in the figure represents that it is a factor of the number just below them. Find and list how many factors are there between numbers 1-20 using the figure below.



Look at the table you have made and give the answers to the questions below.

- 1) List the numbers which have the factor itself and 1?
- 2) What can you say about the numbers which have been underlined?

- 3) List and discuss the numbers between 1-20 which have factors other than itself and 1.

Numbers multiple of itself and 1 is called **Prime number**. The number multiple of itself, 1 and also other numbers is called **Composite number**. A Prime number is only multiple of itself and 1. There are 8 prime numbers between 1-20; 2,3,5,7,11,13,17,19. 1 is neither prime number nor composite number. 2 is the only even prime number. All other prime numbers are odd.

Exercise 2.3:

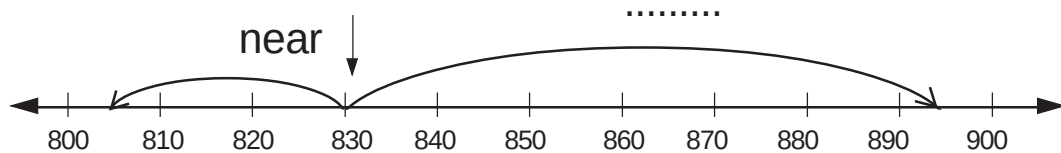
1. A) Write all the numbers between 1-100 in 10x10 boxed lined paper.
B) Cut number 1.
C) Leaving number 2 cut all the numbers divisible by 2.
D) Leaving number 3 cut all the numbers divisible by 3.
E) Leaving number 5 cut all the numbers divisible by 5.
F) Leaving number 7 cut all the numbers divisible by 7.
G) What are the numbers left after finished cutting prime or composite?
H) How many prime numbers are there between 1-100?
I) How many prime numbers are there which are greater than 50 and less than 60?

Note: the method by which we find prime number like we did in question number one is called "eratoesthiniko chalna". It is named after eratoesthini who first derived this method.

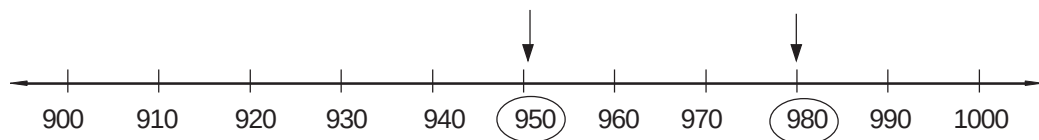
- 2) Find out which numbers given below are prime or composite:
- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| A) 2 | B) 5 | C) 6 | D) 7 | E) 13 | F) 23 |
| G) 33 | H) 41 | I) 53 | J) 72 | K) 75 | L) 79 |
| M) 81 | N) 83 | O) 9 | | | |
- 3) Write all the prime numbers between 75 and 85.
- 4) $11 \times 11 = 121$, 11 is a prime number. What is 121?

2.4 Rounding off the numbers

Look and discuss



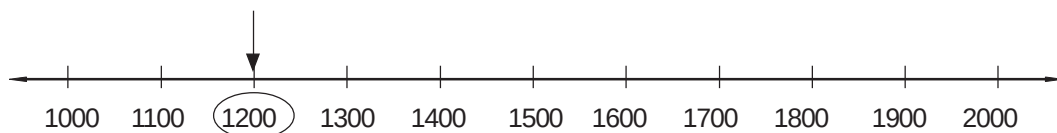
In number line 800 is near to 830 but far from 900. So 800 is the near round off of 830.



In the number line 1000 is near to 980 but far from 900. So the near rounding off of 980 is 1000. Similarly both 1000 and 900 are equidistant from 950 for which near round off of 950 will be 1000.

Example1:

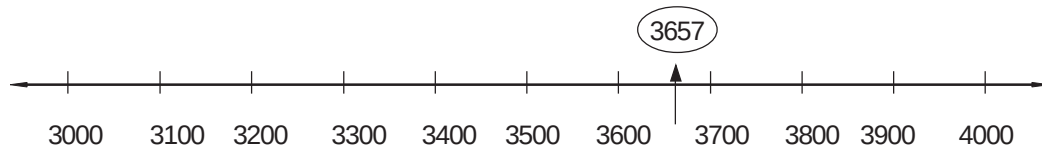
Round off 1200 into nearest thousand.



1000 is nearest to 1200. Hence nearest thousand round off of 1200 is 1000.

Example2:

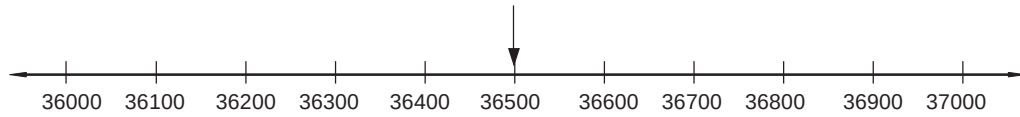
Round off 3657 into nearest thousand.



3657 is nearest to 4000. Hence nearest thousand round off of 3657 is 4000.

Example3:

Round off 36500 into nearest thousand.



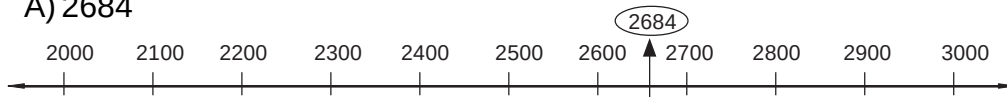
36500 lies just in between 36000 and 37000. The nearest thousand round off of 36500 is 37000.

Teachers note: using number lines discuss about the nearest hundred and thousand round off.

Exercise 2.4

1) Find nearest hundred round off of the given numbers:

A) 2684



B) 3824,

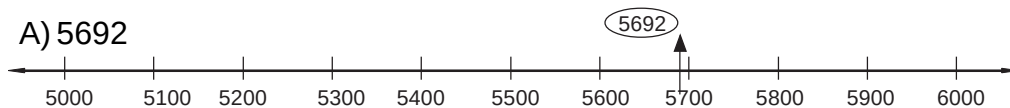
C) 5613,

D) 6123,

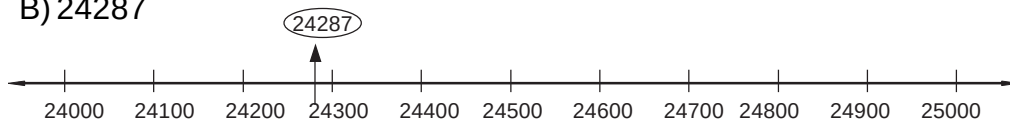
E) 12345

2) Find nearest thousand round off of the given numbers:

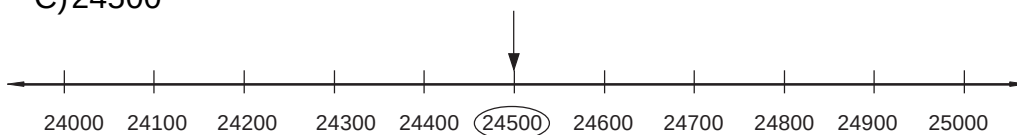
A) 5692



B) 24287



C) 24500



D) 1697, E) 15287, F) 58999, G) 93902

2.5 Square numbers and cubic numbers

2.5.1 Square numbers

Read and discuss the table below. (table from the book)

x	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	6	12	18	24	30	36	42	48	54	60
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

What can we say about the numbers circled in the table?

$$1 \times 1 = 1$$

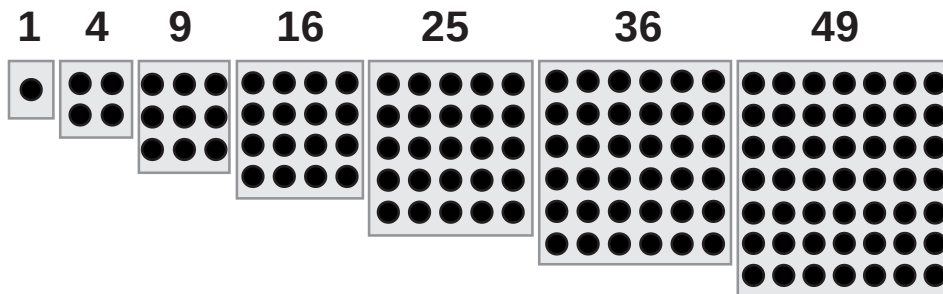
$$2 \times 2 = 4$$

$$3 \times 3 = 9$$

Are the circled numbers multiple of same two numbers?

Can all the circled numbers be expressed the same way? Multiply and see.

Now, draw the circled numbers in a dotted pattern in a box and see what happens?



Here every dotted pattern makes a square. Thus all numbers which represent a dotted square are called a square numbers.

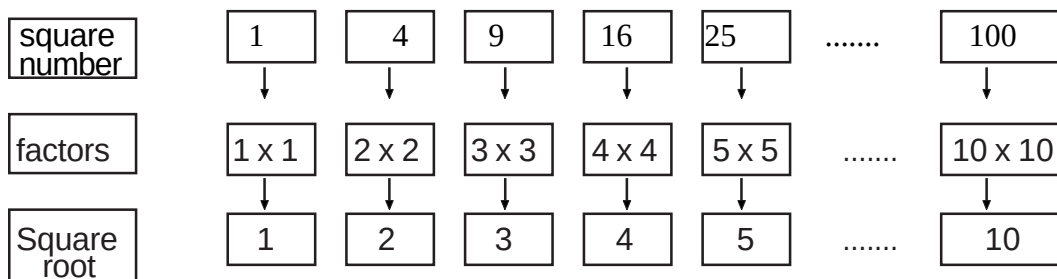
These square numbers can be found by multiplying the number by itself. Look at the table below:

Number	Multiply (number x number)	
1	1×1	1
2	2×2	4
3	3×3	9
4	4×4	16
5	5×5	25

Thus a square number is the product of a number multiplied by itself. So to find out whether a number is a square number or not we have to see whether the number can be expressed or not in the multiplication of two similar numbers. For example $64 = 8 \times 8$ so 64 is a square number but $15 = 3 \times 5$ so 15 is not a square number.

Here, two similar factors of 64 are $8 \times 8 = 64$. 8 is the square root of 64 and 64 is the square number of 8.

Look at the flowchart below to understand square number and square root:



In the above chart when we see from top to bottom it gives square numbers. Square of 3 is 9. When we see from bottom to top it gives square root. Square root of 16 is 4. Similarly square root of 100 is 10.

Example1:

Find square number of 9.

Here,

Given number = 9.

Hence square number = $9 \times 9 = 81$.

Example2:

Chessboard is square in shape and has 64 small square boxes. How many square boxes are there in each rows and column?

Here,

Picture of page 29

Square boxes in chessboard = 64.

$64 = 8 \times 8$.

Hence in each row and column square boxes = 8.

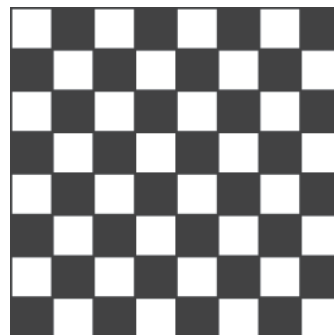
Example 3

Find square root of 25.

Here,

$25 = 5 \times 5$.

Hence, square root of 25 = 5.

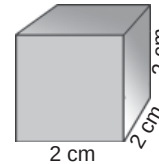


Exercise 2.5.1

- 1) Find square numbers of given numbers:
(a) 2 (b) 5 (c) 6 (d) 7 (e) 8 (f) 9
(g) 10 (h) 11 (i) 12 (j) 13 (k) 14 (l) 15
- 2) Find which of the given numbers are square numbers:
(a) 16 (b) 25 (c) 42 (d) 49 (e) 60 (f) 81
(g) 100 (h) 120 (i) 121 (j) 225
- 3) Find square root of given numbers:
(a) 36 (b) 49 (c) 64 (d) 81 (e) 100 (f) 121
(g) 144 (h) 169 (i) 196 (j) 225
- 4) How many seedlings of cauliflowers are required to plant in an square field where in each line 20 seedlings can be planted?
- 5) In a class 35 students are seated in a square pattern then how many students are there in each line?
- 6) If 144 soldiers are kept in a square pattern during parading, how many soldiers are there in each line?
- 7) When a line is maintained using 45 students in a square pattern, 6 students was not required, what is the total number of students?
- 8) From 1-100 how many square numbers are there?

2.5.2 Cubic Numbers

A dice has length, breadth and height of each 2/2cm. It is a cubical object. Multiple of its length, breadth and height gives; $2 \times 2 \times 2 = 8$. Here there are three sides of 8cm. these numbers are called cubic numbers. In table below, cubic numbers of 1- 5 are given.



Numbers	Product multiplied by three times	Cubic numbers
1	$1 \times 1 \times 1$	1
2	$2 \times 2 \times 2$	8
3	$3 \times 3 \times 3$	27
4	$4 \times 4 \times 4$	64
5	$5 \times 5 \times 5$	125

Hence , to find out the cubic numbers of a given number we should multiply the given number by two similar numbers.

Example 1

Find the cubic number of six.

Here,

Cubic number of 6

$$= 6 \times 6 \times 6$$

$$= 36 \times 6 = 216$$

Example 2

Find cubic root of 64?

Here,

$$64 = 4 \times 4 \times 4.$$

Hence cubic root of 64 = 4.

Exercise 2.5.2

1. Find the cubic numbers of given numbers:

(a) 2

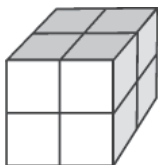
(b) 5

(c) 7

(d) 8

(e) 10

2.



This is a volumetric object with length, breadth and height of 3cm. when it is cut so that its length, breadth and height are 1/1 how many pieces will it be?

3. If one of the three similar sides of an object is 5 then what is the cubic number?

4. Which of the numbers are cubic numbers?

2.6 Prime factorization of numbers

Look at the example below:

$12 = 12 \times 1$ two multiple factorization (12 is the composite number)
 $= 6 \times 2$ two multiple factorization (6 is the composite number)
 $= 4 \times 3$ two multiple factorization (4 is the composite number)
 $= 2 \times 2 \times 3$ three multiple factorization (all are prime factorization)

Can these numbers be more factorised? Discuss.

This process of multiplying a number multiplying it by two similar numbers is called factorization. The process of expressing number in the form of prime numbers is called prime factorization.

Prime factorization can be done in two methods:

1. Continuous division process

Example 1

What is the prime factorization of 12?

Here,

$2 \overline{) 12} \rightarrow$ 12 is even so divided by 2
 $2 \overline{) 6} \rightarrow$ 6 is even so divided by 2
 $3 \rightarrow$ 3 is a prime number so cannot be divided.

$12 = 2 \times 2 \times 3$
 Hence, 12 = 2 x 2 x 3

Hence, $12 = 2 \times 2 \times 3$

Example 2

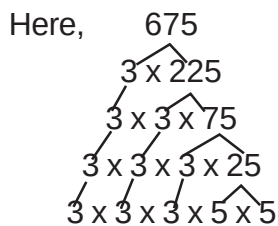
What is the prime factorization of 675?

Here, 3	675	675 was divided by 3
3	225	225 was divided by 3
3	75	75 was divided by 3
5	25	25 was divided by 5
5		Hence, $675 = 3 \times 3 \times 3 \times 5 \times 5$

B) Factor tree method

Example 3

Factorize 675 by factor tree method.



Hence, $675 = 3 \times 3 \times 3 \times 5 \times 5$

Exercise 2.6

1. Using continuous factorization method find the prime factors of numbers below:

- | | | | | |
|--------|--------|--------|---------|---------|
| (a) 8 | (b) 12 | (c) 18 | (d) 24 | (e) 42 |
| (f) 62 | (g) 64 | (h) 85 | (i) 121 | (j) 144 |

2. Use factor tree method to find the prime factors of the numbers below:

- | | | | |
|--------|--------|--------|--------|
| a) 16 | b) 30 | c) 44 | d) 70 |
| e) 162 | f) 252 | g) 676 | h) 625 |
| i) 750 | j) 925 | | |

Lesson 3 Fundamental Operation in Mathematics

3.1 Simplifications for fundamental operations of a problem

Read, discuss and learn

Solution of problems using four basic rules of mathematics; addition (+), subtraction (-), multiplication (x) and division (/) is called simplification. During simplification multiplication or division whichever comes first can be operated.

Then addition or subtraction which ever comes first can be operated.

Example1:

Simplify : $15 \div 5 \times 3 + 7 - 15$

This problem when expressed in words represents; 5 parts of 15 is multiplied by 3 which is added with 7 and 15 is subtracted from it becomes easier to simplify with order of division, multiplication, addition and subtraction.

Hence,

$$\begin{aligned} &15 \div 5 \times 3 + 7 - 15 \\ &= 3 \times 3 + 7 - 15 && (15/3=3, \text{ division}) \\ &= 9 + 7 - 15 && (3 \times 3=9, \text{ multiplication}) \\ &= 16 - 15 && (9+7=16, \text{ addition}) \\ &= 1 && (16-15=1, \text{ subtraction}) \end{aligned}$$

Example 2

Simplify: $55 - 576 \div 12 + 11 \times 3$

Here, $55 - 576 \div 12 + 11 \times 3$

$$\begin{aligned} &= 55 - 48 + 11 \times 3 && [\text{by dividing}] \\ &= 55 - 48 + 33 && [\text{by multiplying}] \\ &= 7 + 33 && [\text{by subtraction}] \\ &= 40 && [\text{by addition}] \end{aligned}$$

Example 3

How much when Twice the 16 is subtracted by 20 and 13 is added to it?
When this is represented in mathematical form it becomes $16 \times 2 - 20 + 13$.

$$\begin{aligned}\text{Hence, } 16 \times 2 - 20 + 13 \\ &= 32 - 20 + 13 && [\text{by multiplying}] \\ &= 12 + 13 && [\text{by subtraction}] \\ &= 25 && [\text{by addition}]\end{aligned}$$

$$\begin{aligned}\text{Or,} \\ 16 \times 2 - 20 + 13 \\ &= 32 - 20 + 13 \\ &= 32 + 13 - 20 \\ &= 45 - 20 \\ &= 25\end{aligned}$$

Exercise 3.1

Simplify:

- | | |
|---|---|
| (1) $44 + 24 \div 3 - 30$ | (2) $63 \div 9 \times 7 + 4 - 52$ |
| (3) $6 \times 64 \div 16 + 7 - 21$ | (4) $24 \times 12 \div 12 - 24 + 17$ |
| (5) $55 \div 11 + 7 \times 3 - 13$ | (6) $132 \div 12 \times 12 - 124 \div 31$ |
| (7) $422 + 124 \div 4 \times 2 - 355$ | (8) $144 \div 24 - 3 \times 15 \div 5 + 16$ |
| (9) $625 \div 25 - 25 \times 25 \div 5 + 100$ | (10) $576 \div 24 + 51 \div 17 - 20$ |

(11) Convert all the word form of problem into mathematical form:

- A) How much is five times two is added by 3?
- B) How much is 12 times 3 added by 7?
- C) How much is one third of 36 subtracted by 5 and added by 7?
- D) How much is 10 times 15 added by 50 and 200 is subtracted to it?
- E) How much is when thrice of one third of 9 is subtracted from 9?

3.2 Uses of brackets in Simplification

Read, discuss and learn

How much will it cost to buy copy of Rs.10 eraser of Rs.2 for three people?

Cost for one person = $10 + 2$

Cost for three people = $3 \times (10 + 2)$

= $3 \times (10 + 2)$ simplifying

= 3×12 [operation inside brackets]

= 36 Rs

Example 1

Simplify : $12 - (20 - 12)$

Here, $12 - (20 - 12)$

= $12 - 8$ [operation inside bracket]

= 4

Example 2

Simplify: $5 + 8 \times 2 - (5 - 2)$

Here, $5 + 8 \times 2 - (5 - 2)$

= $5 + 8 \times 2 - 3$ [operation inside bracket]

= $5 + 16 - 3$ [multiplication]

= $21 - 3$ (addition)

= 8 (subtraction)

Example 3

How much is it when we add 2 to 5 multiply it by 3 and subtract 6 from it ? Addition, subtraction and multiplication are involved in this problem. Among these we should do multiplication first. But according to the problem it is three times the total of 5 and 2. So we should do addition first.

Hence , addition of 2 to 5 comes first so the mathematical expression using bracket is:

$$(5+2) \times 3 - 6$$

$$= 7 \times 3 - 6 \quad \text{\{first simplification is done inside bracket. After that}$$

$$= 21 - 6 \quad \text{bracket should be removed\}}$$

$$= 15$$

Therefore if there is bracket in the problem then simplification should be first performed inside the bracket and then remaining simplification should be performed. Simplification in brackets should be done according to order () small brackets, {} middle brackets and then [] larger brackets.

Example 4

Simplify: $8 + 14 \times \{(8 - 2) + 3\} \div 18$

$$\text{Here, } 8 + 14 \times \{(8 - 2) + 3\} \div 18$$

$$= 8 + 14 \times \{6 + 3\} \div 18$$

$$= 8 + 14 \times 9 \div 18$$

$$= 8 + 14 \times \frac{9}{18}$$

$$= 8 + 7$$

$$= 15$$

Or,

$$8 + 14 \times \{(8 - 2) + 3\} \div 18$$

$$= 8 + 14 \times \{6 + 3\} \div 18$$

$$= 8 + 14 \times 9 \div 18$$

$$= 8 + 126 \div 18$$

$$= 8 + 7$$

$$= 15$$

Exercise 3.2

(A) Simplify:

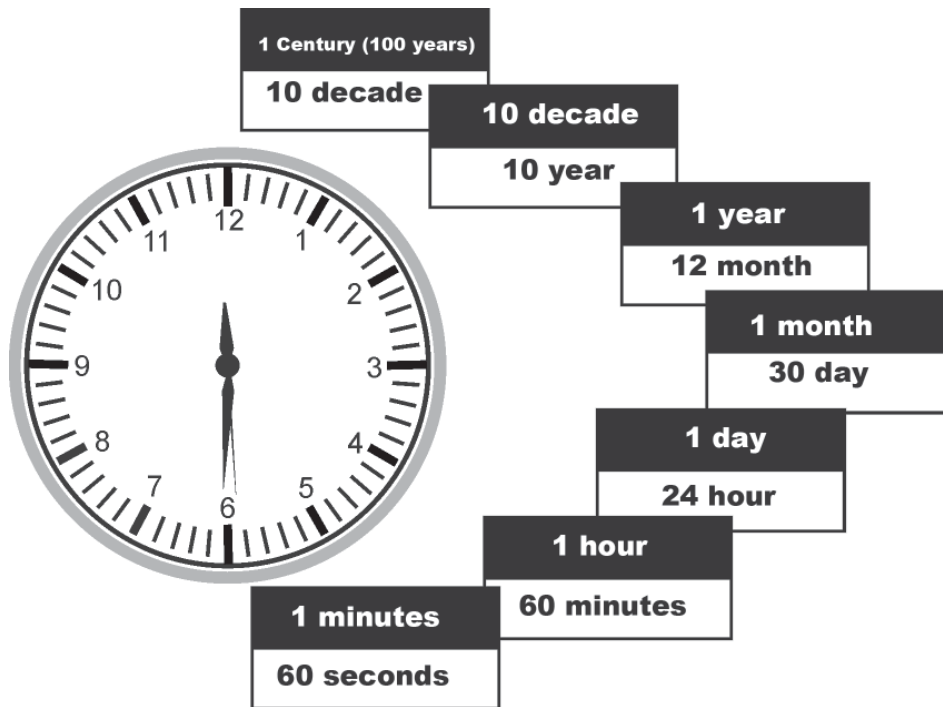
- (1) $6 - (5 - 2)$
- (2) $(16 - 4) \times (5 - 3)$
- (3) $(16 + 4) + 5 - 3$
- (4) $20 - \{8 - (5 + 2)\}$
- (5) $3 - \{12 \div (2 \times 3)\}$
- (6) $19 - 7 + \{4 - (5 - 2)\} \times 2$
- (7) $3 + 4 \div \{2 + 4 \div (4 - 2)\}$
- (8) $30 \div \{60 - 3(21 - 6)\}$
- (9) $22 \div \{20 \div (4 + 6)\} \times 2$
- (10) $80 - 5 \{9 - (14 - 12)\} \div 5$

2) Simplify by converting into mathematical form:

- (1) How much is thrice the difference between 12 and 5 divided by 7?
- (2) How much is when thrice the difference between 9 and 5 is subtracted from 20?
- (3) How much is when difference between 70 and 30 is subtracted from five times the sum of 7 and 3?
- (4) How much is when difference between 16 and 7 is subtracted from one fourth the sum of 16 and 20?
- (5) How much is when one third of two times 12 is subtracted from 50 and is divided into 6 parts?
- (6) How much is when 4 added to difference between 5 and 2 is subtracted from 7 and 9 is added?

Lesson 4

Time



Compare the units of time given above with each other and find out their relationship. The small hand (hand showing seconds) of the clock crosses sixty smaller lines when it makes a circle around the clock. At that time the long hand (hand showing minute) of the clock reaches just another line. Now discuss with your friends and find out answers to the following questions which are related to the speed of the hands of the clock and their relationship with each other:

- How many lines does a minute hand cross while the second hand makes a full circle?
- How many full circles does a second hand make while the minute hand makes one full circle?
- How much time does the hour hand take to make one full circle around the clock?
- How many circles does an hour hand make in a day? And how many circles does a minute hand make around the clock in a day?

Example 1

Multiply 3 years, 8 months and 12 days by 3:

Multiplying years, months and days separately by 3,

Year	Months	Day
3	8	12
	x	3
9	24	36
9	25	6
11	1	6

30 days = 1 month, 36 days = 1 month 6 days, 12 months
= 1 year, 25 months = 2 years 1 month

Example 2

Divide 15 hours 30 minutes 48 seconds by 4.

Dividing hours, minutes and seconds separately by 4.

3 H. 52 M. 42 S.

4	)	15	30	48
		-12		
		3	30	48
		210	48	
		-20		
		10		
		-8		
		2	48	
		168		
		-16		
		8		
		-8		
		0		

30 hours = 180 minutes
2 minutes = 120

Example 3

Divide 10 years 7 months 12 days by 6:

Dividing years months and days separately.

$$\begin{array}{r}
 \text{1 Yr. 9 Mon. 7 Day} \\
 \hline
 6 \overline{) \begin{array}{r} \text{Yr. Mon. Day} \\ 10 \quad 7 \quad 12 \\ \underline{6} \\ 4 \quad 7 \quad 12 \end{array}} \\
 \hline
 \begin{array}{r} 55 \quad 12 \longrightarrow 4 \text{ years} = 48 \text{ months} \\ \underline{54} \\ 1 \quad 12 \\ \underline{42} \longrightarrow 1 \text{ month} = 30 \text{ days} \\ 42 \\ \underline{} \\ \text{x} \end{array}
 \end{array}$$

Example 4

A micro bus travelled to Kathmandu from Narayangard continuously for 6 times. If the bus took 28 hours and 34 minutes for completing its journey 6 took and if it takes the same amount of time for each journey. How much time did it take for its one journey from Narayangard to Kathmandu?

Total time taken for 6 journeys = 28 hours & 34 minutes

So time taken for 1 journey is,

$$\begin{array}{r}
 \text{4hours 45minutes 40 seconds} \\
 6 \overline{) \begin{array}{r} 28 \quad \quad 34 \\ \underline{-24} \\ 4 \times 60 \longrightarrow 240 + 34 \\ \underline{} \\ 274 \\ \underline{-24} \\ 34 \\ \underline{-30} \\ 4 \times 60 \quad - 240 \\ \underline{} \\ 4 \times 60 \quad - 240 \\ \underline{} \\ 0 \end{array}}
 \end{array}$$

Exercise 4

1. Multiply:

- a. 2 years 7 months 16 days by 7
- b. 5 years 6 months 12 days by 5
- c. 7 years 4 months 18 days by 3
- d. 8 years 9 months 6 days by 8
- e. 6 hours 40 minutes 15 seconds by 5

2. Divide:

- a. 10 years 8 months 20 days by 2
 - b. 15 years 6 months 21 days by 3
 - c. 13 years 7 months 12 days by 6
 - d. 5 hours 7 minutes 32 seconds by 4
 - e. 8 hours 10 minutes 35 seconds by 7
3. If it takes 6 hours and 30 minutes to do half of the work, how much time would it take to do the whole work?
4. If it takes 2 hours and 15 minutes to plant the seedlings of cauliflower in a square-shaped field. How much time would it take to do the same in six such fields?
5. A pipe fills up a tank in 1 and $\frac{1}{2}$ hours. How much time would the same pipe take to fill up 5 tanks of the same size?
6. Seven subjects are taught in Shila's class. If it takes 45 minutes for one subject, how many minutes would Shila take to study all 7 subjects?
7. Shiva finishes up the homework of all 7 subjects in 3 and $\frac{1}{2}$ hours. If he spends equal time for doing each subject's homework then how much time does he take to do homework of 1 subject?

8. The time table of the departure of buses from Pokhara is given below. Both the buses A and B departed from Pokhara to Kathmandu at 5:30 and reached the following places at the given times:

Pokhara	Bus A	Bus B
Khairaitar	6:45 am	7:00 am
Damauli	7:50 am	8:45 am
Dumre	9:30 am	10:30 am
Anbukhairani	10:15 am	12:30 noon
Muglin	11:00 am	1:00 pm
Mahadev Besi	11:50 am	2:30 pm
Naubise	1:00 pm	4:00 pm
Kathmandu	2:00 pm	5:30 pm

Now answer the following questions:

- At what time did Bus A reach Damauli?
- How long after Bus A did Bus B reach Damauli?
- How long will it take for passengers from Dumre to reach Muglin if they are in Bus A?
- How long did the Bus B take to reach Mahadev Besi from Dumre?
- Where was Bus B while Bus A was at Naubise?
- Which bus shall we take to reach Mahadev Besi within 2 pm?
- How long after Bus A does Bus B reach Kathmandu?

Lesson 5

MONEY

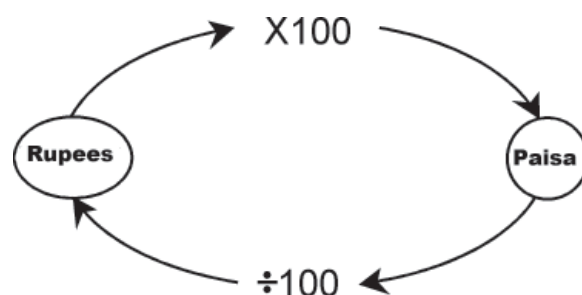
We can use coins and notes and make payments in different ways. For example to make a payment of 1 rupee 25 paisa,

- Note of Rs 1 and 1 coin of 25 paisa
- Two coins of 50 paisa and 1 coin of 25 paisa
- Five coins of 25 paisa.... Etc

These are not only ways to make payment 1 rupee 25 paisa. Discuss in class and find out into how many ways we can make payment of 1 rupee 25 paisa. To denote 1 rupee 25 paisa we write Rs 1.25. We put a decimal between rupee and paisa, why?

Rs 1 has hundred paisa or 100 paisa = Rupee 1. So to convert a rupee into paisa we should multiply it by 100. Similarly, because 100 paisa denotes Rs 1 we should divide the rupee by 100 to convert it into rupee.

Try to understand this fact from the following circle:



According to the example above there is 125 paisa in 1 rupee 25 paisa and to convert 125 paisa into rupee we should divide it by 100 which give us Rs 1.25.

Example 1

Convert Rs 4.35 into paisa:

Here, Rs 4.35 = $(4 \times 100 + 35)$ paisa = 435 paisa.

Example 2

Convert 525 paisa into rupees:

525 paisa = Rs $(525 \div 100)$ = Rs 5.25.

Example 3

If Ramesh gives Rs 20 note to the shopkeeper to make a payment of Rs 10.75 how much change will he get back?

Here, Rs 20 = 20×100 = 2000 paisa.

And Rs. 10.75 = $10 \times 100 + 75$ p = 1075 p

So the amount he will get back = $(2000 - 1075)$ p = 925 p = Rs 9.25.

(To convert paisa into rupee we put decimal before two numbers from the right)

Subtraction in short,

$$\begin{array}{r} \text{Rs } 20.00 \\ - \text{Rs } 10.75 \\ \hline \text{Rs } 9.25 \end{array}$$

Alternatively: (Subtracting rupee from rupee and paisa from paisa)

Rs	P		Rs	P
20	00	=	19	100
10	75		- 10	75
10	25		Rs 9	25

In this way to subtract or add rupee and paisa we add or subtract rupee with rupee and paisa with paisa.

Example 4

(a) How much is the total of Rs 6.80 and Rs 7.25?

Adding rupee and paise separately,

Rs	P
6	80
+7	25
13	105
	140 5

{ 105 paise is equal to 1 rupee 5 paise. So Rs 1 added to Rs 13 the total becomes Rs 14.05.

= Rs. 14.05

In short,

$$\begin{array}{r} \text{Rs. 6.80} \\ + \text{Rs. 7.25} \\ \hline = \text{Rs. 14.05} \end{array}$$

(b) How much will it be if we subtract 3 Rs 75 paise from 5 Rs 25 paise?

Subtracting rupee and paise separately,

Rs	P
4 5	25 ¹²⁵
1	50
= ?	1.50

We can not subtract 75 p from 25p. So we borrow Rs 1 = 100p.

Picture of page 46

In short,

$$\begin{array}{r} \text{Rs } 5.25 \\ - \text{Rs } 3.75 \\ \hline \text{Rs } 1.50 \end{array}$$

This is the subtraction of decimal



Example 5

How much does Kamalesh earn in 7 days if he earns Rs 25.35 per day?

Here, to calculate Kamalesh's 7 days income we should add his per day income of Rs 25.35 seven times. We can calculate it in short. Therefore income of Kamalesh in 7 days = Rs 25.35.

x 7

Rs 177.45

Another method:

Multiplying rupee and paisa separately,

Rs	P	
25	35	
x 7		
175	245	Since 245p = Rs 2.45 the total is Rs 177.45
177	45	

Therefore Kamalesh's income is Rs 177.45.

Example 6

If father gave his son Nima Rs 125 as pocket money for 10 days then how much money was given to him per day?

Here, Rs 125 has to be spent in 10 days. So we should divide Rs 125 in 10 equal parts. So this is a problem of division.

Now dividing Rs. 125 by 10,

$$\begin{array}{r}
 \text{Rs. 12.50} \\
 10 \overline{) \text{Rs. 125}} \\
 \underline{- 10} \\
 25 \\
 \underline{- 20} \\
 5 \times 100 \text{ p} \\
 \underline{500 \text{ p}} \\
 - 50 \\
 \hline
 0
 \end{array}$$

Converting the remainder Rs. 5 into paisa, the total becomes 500 paisa.

Hence, per day pocket money = Rs. 12.50

Next method;

$$\begin{array}{r}
 12 \\
 10 \overline{) \text{Rs. 125}} \\
 \underline{- 10} \\
 25 \\
 \underline{- 20} \\
 5
 \end{array}$$

Hence per day pocket money = Rs. $12\frac{5}{10}$ = Rs. 12.50

Example 7

If you buy 12 pencils at Rs. 2.15 per piece and sell them at Rs. 2.50 per piece how much will be the profit?

Here,

Selling price of pencils per piece = Rs 2.50

Cost price of pencils per piece = Rs 2.15

Profit per pencil = Selling price – Cost price

$$= \text{Rs } 2.50 - \text{Rs } 2.15$$

$$= \text{Rs } 0.35$$

Profit of 12 pencils = Rs 0.35 x 12 = Rs 4.20

Hence profit = Rs 4.20

Exercise 5

1. How many coins of the following denominations will be there in a rupee?

- A) 25 paisa B) 50 paisa C) 10 paisa

2. Convert into paisa:

- A) Rs. $\frac{1}{2}$ B) Rs $\frac{1}{4}$ C) Rs $\frac{1}{10}$ D) Rs $\frac{1}{5}$
E) Rs $1\frac{1}{2}$ F) Rs 5.05 G) Rs 10.15 H) Rs 75.35

3. Convert into rupees:

- A) 125 paisa B) 105 paisa C) 325 paisa D) 10 paisa
E) 15 paisa F) 20 paisa G) 25 paisa H) 50 paisa
I) 1235 paisa J) 2375 paisa K) 5437 paisa L) 4857 paisa

4. Look at the price list and find out the total cost of the following particulars:

Price List	
Particular	Price
Pencil	Rs. 1.50 ,
Copy	Rs. 3.50 ,
Book	Rs. 18.75 ,
Geometry Box	Rs. 45.00 ,
Bag	Rs. 150.00
Pen	Rs. 10.00

- a. 1pencil, 1 pen and 1 copy
b. 1 bag and 1 geometry box
c. 1 book and 1 pencil
d. 5 copies
e. 5 books and 10 copies
f. 2 pens and 4 copies

- 5. Find out the difference of cost price of the following objects on the basis of the cost price as given in question 4 above:**
- A) Pencil and pen B) Bag and geometry box
C) Book and copy D) Pencil and geometry box
- 6. The cost of three packets of biscuits and four eggs is Rs 40. If the cost of 3 packets of biscuits is Rs 27.**
- A) What is the cost of 4 eggs?
B) What is the cost of a packet of biscuit?
- 7. What is the cost of a battery at the rate of Rs10 per piece?**
- 8. The fare of boating in fewa lake is Rs 15 per person. The fare of children is just the half of adult's rate. How much will Mrs Bimala and her 3 children have to pay for boating?**
- 9. If you buy good of Rs 225.65 and give a Rs 500 note to the shopkeeper, how much will he return?**
- 10. Pawan gave Rs 53.75 to his children as pocket money. If equal amount of pocket money was given to each of them, find out the amount each child was given?**

Lesson 6

DISTANCE

Division and multiplication of unit of distance

Guess, what will be the distance between your house and school? What is the distance you will have to walk in a week from home to school? How can we find it out? Discuss.

Example1:

Multiply 3km, 200m, 55cm by 8:

Multiplying the units of distance separately,

Km	m	cm	
3	200	55	
		x 8	
24	1600	440	440cm = 4m 40 cm
24	1604	40	1604m = 1km 604cm
25	604	40	

Example2:

Divide 5km, 600m 56cm by 8:

Here, dividing the units of distance separately,

	0km	700m	7cm	
8)	5	600	56	
	0	5000		← 5x1000
		5600	56	
		- 56		
		00	56	
			- 56	
			x	

Hence, dividend = 700m 7cm.

Example3:

Hari's school is 1km 5m far from his home. If the distance of Shyam's school is 3 times more than Hari's what is the distance between Shyam's home and school?

Here, the distance between Hari's home and school= 1km 5m

The distance between Shyam's home and school

= 3 x (distance between haris home and school)

$$= 3 \times (1\text{km } 5\text{m})$$

$$= 3\text{km } 1500\text{m}$$

$$= 4\text{km } 500\text{m}$$

Hence, the distance between shyam's home and school= 4km 500m

Example 4:

If there is a plan to blacktop a road of 35km 600m 80cm length, dividing it into 4 equal sections. What is the length of 1 section?

Here, division of 55km 600m 80 cm road into 4 equal sections means dividing the distance by 4. Therefore dividing the units of distance seperately by 4,

	13km	900m	200cm	
4)	km		m	cm
	55	600	80	
	- 4			

	15			
	- 12			

	3	600	80	
		3600	80	
		- 36		

		00	80	
			- 8	

			0	

Hence the length of each section blacktopped = 13km 900m 20cm

Exercise 6

1. **Guess the length of the following objects and measure them, were you correct?**

- The length and breadth of “My Mathematics Class 5”.
- The length and breadth of the blackboard.
- The length and breadth of your book.
- The length and breadth of your classroom door.
- The length and breadth of your classroom.

2. **Convert the following units of distance into cm:**

- 1km 75cm
- 5m 65cm
- 1km 200m 80cm
- $1\frac{1}{2}$ m
- 5km 350m 75cm
- 3.2m
- 5.72m

3. **Convert the following units of distance into m.**

- 1km 200m
- 1.5km
- 3m 80cm
- 5m 70cm
- 250cm
- 15cm

4. **Multiply:**

a. km m cm

5 320 25

x6

b. km m cm

3 750 75

x9

$$\begin{array}{r}
 \text{c. km} \quad \text{m} \quad \text{cm} \\
 6 \quad 20 \quad 5 \\
 \text{x12} \\
 \hline
 \end{array}$$

$$\begin{array}{r}
 \text{d. km} \quad \text{m} \quad \text{cm} \\
 8 \quad 425 \quad 60 \\
 \text{x15} \\
 \hline
 \end{array}$$

$$\begin{array}{r}
 \text{e. km} \quad \text{m} \quad \text{cm} \\
 17 \quad 250 \quad 65 \\
 \text{x17} \\
 \hline
 \end{array}$$

$$\begin{array}{r}
 \text{f. km} \quad \text{m} \quad \text{cm} \\
 22 \quad 560 \quad 30 \\
 \text{x25} \\
 \hline
 \end{array}$$

5. Divide:

- 5km 600m 75cm by 5.
 - 3km 145m 20cm by 6.
 - 12km 200m 64cm by 8
 - 8km 650m 71cm by 9.
 - 12km 330m 96cm by 6.
 - 2km 580m 83cm by 7.
 - 3.693km by 3.
- If the height of a brick is 12cm. how many bricks do you need to make a wall of 3.6m height? (cement not counted)
 - If a 6m long ribbon is equally divided among 6 girls. What will be the length of ribbon each girls get?
 - If a 2m 50cm sugar cane was eaten up by 5 persons in equal poportion. What length did each get?
 - A 200km long road was balck topped dividing it into 8 equal sections. What was the length of each section?
 - At the rate of 10km 650m per month, if a pathis completed in 7 months. What is the length of the road?

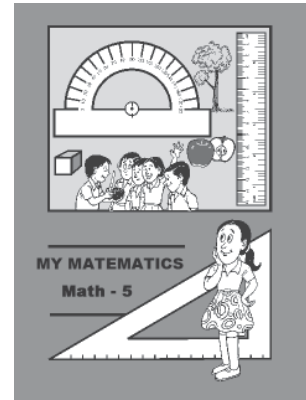
Lesson 7

PERIMETER

Perimeter of Rectangular objects

Lets tie the book of “ Mathematics class 5” from its length side to its every end with a thread. What length of thread we will need to tie the book in this way?

Will a measurement of book of length = 24 cm and breath = 18cm be equal to the thread used to cover its every end?



Measurement of the outer end of any surface or objects is called its perimenter.

How many times do we have to measure the length to find out the perimeter of a rectangular book?

Do we have to measure the breadth same way as length?

Here, perimeter of book “ Mathematics class 5”
 $= 24 \text{ cm} + 18 \text{ cm} + 24 \text{ cm} + 18 \text{ cm} = 84 \text{ cm}.$

While calculating Perimeter of any Rectangular surface or object,

$$\begin{aligned}\text{Perimeter (p)} &= \text{length} + \text{breadth} + \text{length} + \text{breadth} \\ &= 2 \times \text{length} + 2 \times \text{breadth} \\ &= 2 (\text{length} + \text{breadth}) \\ &= 2 (l + b)\end{aligned}$$

When the surface is square,

$$\begin{aligned}\text{Perimeter (p)} &= 2 (\text{length} + \text{breadth}) \\ &= 2 (\text{length} + \text{length}) \text{ because length} = \text{breadth} \\ &= 2 (2 \text{ length}) \\ &= 4 \text{ length} = 4l\end{aligned}$$

So, perimeter of square = $4 \times \text{length}.$

Example 1

What is the perimeter of an object having length = 8 cm and breadth = 6cm?

Here,

Length of rectangle (L) = 8 cm

Breadth (b) = 6 cm

Perimeter (p) = ?

$$\begin{aligned}\text{According to formula, perimeter (P) of rectangle} &= 2 (L + b) \\ &= 2 (8\text{cm} + 6 \text{ cm}) \\ &= 2 \times 14 \text{ cm} \\ &= 28 \text{ cm.}\end{aligned}$$

Therefore the circumference of rectangle = 28 cm.

Example 2

If the length of a square is 8 cm what will be its perimeter?

Here,

Length of the square (L) = 8 cm

Perimeter (p) = ?

$$\begin{aligned}\text{According to formula, perimeter of a square (p)} &= 4 L \\ &= 4 \times 8 \text{ cm} \\ &= 32 \text{ cm}\end{aligned}$$

Therefore the perimeter of the square = 32 cm

Example 3

What length of barbed wire will be required to barbe it around a field 5 times which has length 50 meter and breadth 30 meter?

Length of barbed wire = 5 times of the perimeter of the field.

Here,

length (L) = 50 m

breadth (b) = 30 m

perimeter (p) = ?

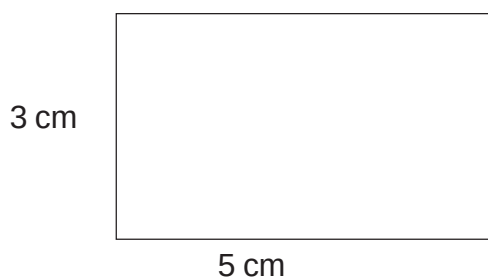
$$\begin{aligned}\text{Perimeter of field (p)} &= 2 (L + b) \\ &= 2 (50\text{m} + 30\text{m}) \\ &= 2 \times 80 \text{ m} \\ &= 160 \text{ m}\end{aligned}$$

Therefore, the length of barbed wire = $5 \times 160 \text{ m} = 800 \text{ m}$.

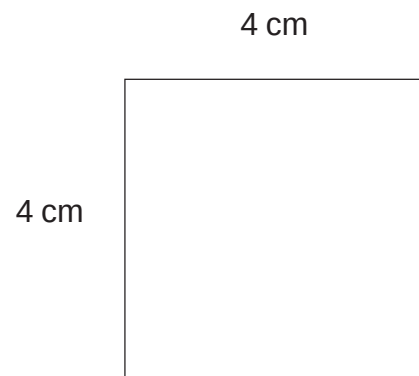
Exercise 7

1. Find out the perimeter of the following shapes:

(a)

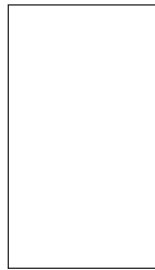


(b)



(c)

3.5 cm



2 cm

(d)

5 cm



5 cm

2. Find out the perimeter of rectangular surface having the following length and breadth:

a. $L = 8\text{ cm}$, $b = 6\text{ cm}$

b. $L = 7\text{ cm}$, $b = 3\text{ cm}$

c. $L = 6\text{ cm}$, $b = 5\text{ cm}$

d. $L = 10\text{ cm}$, $b = 5\text{ cm}$

e. $L = 6.8\text{ cm}$, $b = 3\text{ cm}$

f. $L = 5.6\text{ cm}$, $b = 2\text{ cm}$

g. $L = 10.3\text{ cm}$, $b = 6.5\text{ cm}$.

3. Find out the perimeter of a square with the following length:

a. Length = 3 cm

b. $L = 5\text{ cm}$

c. Length = 8 cm

d. $L = 12\text{ cm}$

e. Length = $5\frac{1}{2}\text{ cm}$

f. $L = 7.5\text{ cm}$

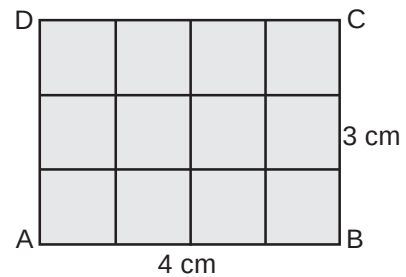
4. What will be the perimeter of a field which is 55 m in length and 40 m in breadth?
5. If a rectangular Chautari is 12 meter in length what will be its perimeter?
6. A square- shaped field is 30m in length. If you put a wall around it how long will the wall be?
7. What length of barbed wire will be needed to fence around a rectangular field with 60m . length and 40m breadth 3 times?

Lesson 8

AREA

Area of a rectangular surface

To say how much is the area of an object means counting how many squares of 1 square unit it contains.



In the picture the length of the rectangle is 4 cm. and its breadth is 3 cm. There are 12 squares of 1 square unit. Therefore the area of the rectangle ABCD is 12 square centimeter. Square centimeter is called cm^2 in short. To count square unit always in every given figure for calculating area is difficult. Therefore to find out the shortcut method read, discuss and complete the table given below.

Rectangular Surface	Length	Breadth	Area (Counting squares)	Length x Breadth
	3 cm	1 cm	3 cm^2	$3 \text{ cm} \times 1 \text{ cm} = 3 \text{ cm}^2$
	3 cm	2 cm	6 cm^2	$3 \text{ cm} \times 2 \text{ cm} = 6 \text{ cm}^2$
	4 cm	2 cm	8 cm^2	
	4 cm	3 cm	12 cm^2	

What did you learn from the table?

Is product of length and breadth of square equal to its area?

From discussion above:

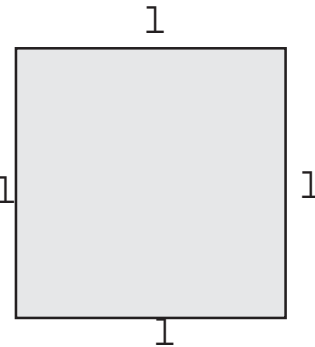
Area of rectangular object = length x breadth

Or $A = L \times b$

A square shaped object has its length and breadth equal.

Therefore $b = L$ and $L \times b = L \times L$

So the area of surface (A) = L^2



Example 1

What is the area of a square which is 5 cm in length and 3 cm in breadth?

Here, Length (L) = 5 cm

Breadth (b) = 3 cm

Area (A) = ?

According to the formula,

$$A = L \times b$$

$$= 5 \text{ cm} \times 3 \text{ cm} = 15 \text{ cm}^2$$

$\therefore$ Area of the rectangle = 15 cm^2

Lets divide the rectangle into 1 square cms:

Now there are 15 squares of 1 square cm.

Therefore area A = 15 cm^2



Example 2

What is area of a square which is 5 cm in length?

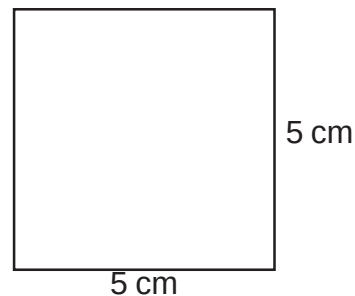
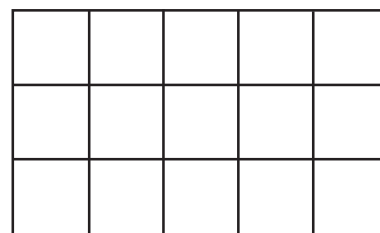
Here, Length (L) = 5 cm

Area (A) = ?

Area of square (A) = L^2

$$= (5 \text{ cm})^2$$

$$= 25 \text{ cm}^2$$



Example 3

If the length of a rectangle room is 6 m and breadth is 4 m, what will be its area?

Here, length of room (L) = 6m

Breadth (b) = 4m

Area (A) = ?

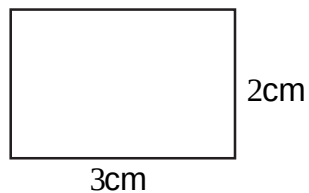
So, area (A) = $L + b$
= $6\text{m} + 4\text{m}$
= 24 m^2

Area of room = 24 m^2

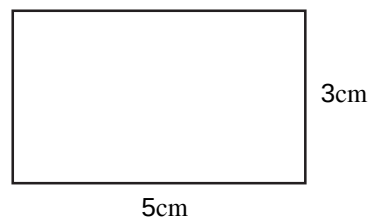
Exercise 8

1. Find out the area of each of the following rectangular shapes:

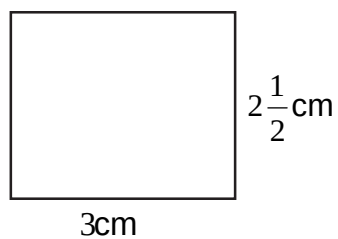
(a)



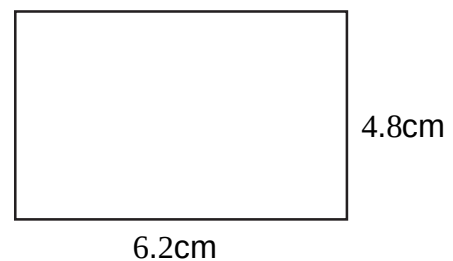
(b)



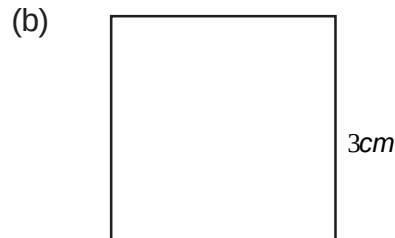
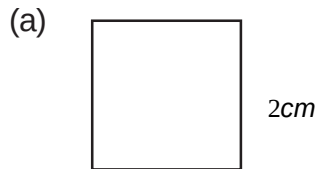
(c)



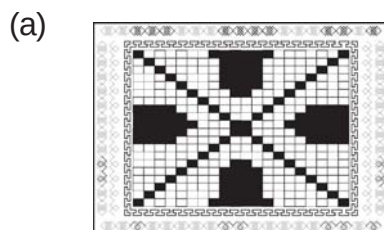
(d)



2. Find out the area of each of the following squares:



3. Find out the area of the surface of each of the following objects:



length of carpet = 2.5 m
breadth = 1.8 m



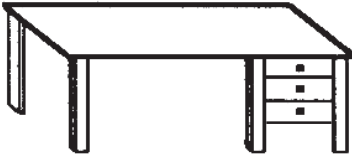
length of frame = 30cm
breadth = 22cm

(c)

length of book = 20.2 cm
breadth = 15.9 cm



(d)



length of the surface of the table = 1.3m
breadth = 80 cm

4. The length of the surface of a table is 3.1 m and its breadth is 24m. What will be its area?
5. What will be the area of a rectangular field of length 52 m and breadth 32.5m?
6. Find the area of a square handkerchief which has length of 30 cm.

Lesson 9

CAPACITY

How much water does a glass contain?

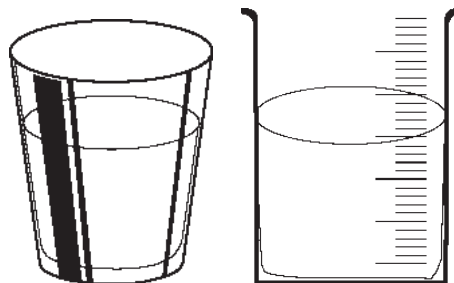
How much water fills up a water pot?

To answer such questions we have to measure the water in it. We use objects like cylinder or litre to measure liquid.

Units like litre and milliliters are used to measure water or other liquids. Capacity is the measurement of the amount of liquid in a container.

Litre and milliliters are the units used to measure capacity.

1 litre (l) = 1000 millilitre (ml)



Example 1

How many milliliters are there in 5 litres and 200ml.

Here, 5 l = 5000ml

= 5 × 5000 ml

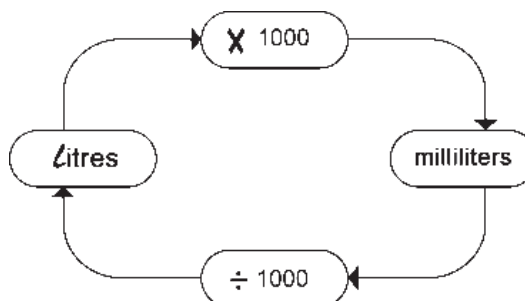
Now, 5l, 200ml = (5000 + 200) ml

= 5200ml

Example 2

Multiply:

$$\begin{array}{r} 1 \text{ L} \\ 2 \text{ 300} \\ \times 4 \\ \hline 8 \text{ 1200} \\ = 9 \text{ 1 200 L} \end{array}$$



Example 3

Solve : $15\cancel{1} 600\text{ m}\cancel{1} \div 5$

$$\begin{array}{r} 3\cancel{1} \quad 120\text{m}\cancel{1} \\ 5 \overline{) \quad 1 \quad \text{m}\cancel{1}} \\ \underline{15} \quad 600 \\ -15 \\ \hline \text{X} \quad 600 \\ \quad \underline{-5} \\ \quad \quad 10 \\ \quad \quad \underline{-10} \\ \quad \quad \quad 0 \end{array}$$

Example 4

Capacity of a glass is $250\text{ m}\cancel{1}$. Sujan drank 5 glasses of water at one time. How much water did he drink? Express in litres.

$$\begin{aligned} \text{Here, 5 glasses} &= 5 \times 250\text{ m}\cancel{1} \\ &= 1250\text{ m}\cancel{1} \\ &= 1250 \div 1000\cancel{1} \\ &= 1.250\text{ litres} \end{aligned}$$

Example 5

How many glasses with the capacity of $500\text{m}\cancel{1}$ are required to fill up a jug with a capacity of $2\cancel{1} 500\text{ m}\cancel{1}$?

Here,

$$\text{Capacity of jug} = 2\cancel{1} 500\text{ m}\cancel{1}$$

$$= (2 \times 1000 + 500 \text{ ml}) \text{ ml}$$

$$= 2500 \text{ ml}$$

Capacity of glass = 500ml

Therefore,

$$\begin{aligned} \text{Number of glasses required to fill up the bottle} &= \frac{2500}{500} \\ &= 5 \end{aligned}$$

∴ The jug will be filled up with 5 glasses.

Exercise 9

(1) Solve:

$$\begin{array}{r} \text{(a)} \quad \text{l} \quad \text{ml} \\ 3 \quad 400 \\ \hline \quad \quad \times 3 \end{array}$$

$$\begin{array}{r} \text{(b)} \quad \text{l} \quad \text{ml} \\ 15 \quad 250 \\ \hline \quad \quad \times 4 \end{array}$$

$$\begin{array}{r} \text{(c)} \quad \text{l} \quad \text{ml} \\ 21 \quad 250 \\ \hline \quad \quad \times 6 \end{array}$$

$$\begin{array}{r} \text{(d)} \quad \text{l} \quad \text{ml} \\ 16 \quad 750 \\ \hline \quad \quad \times 8 \end{array}$$

$$\text{(e)} \quad 10 \text{ l. } 200 \text{ ml} \div 3$$

$$\text{(f)} \quad 15 \text{ l. } 750 \text{ ml} \div 6$$

$$\text{(g)} \quad 48 \text{ l. } 500 \text{ ml} \div 5$$

$$\text{(h)} \quad 28 \text{ l. } 250 \text{ ml} \div 10$$

2. If a glass's capacity is 280 ml of water. How many glasses are required to fill a jug with the capacity of 4 l 200ml?
3. What capacity of kettle is required to serve 300ml of tea per person at once to 50 people?

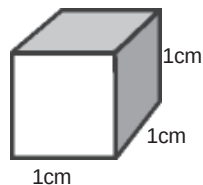
4. With a pot of 750 ml capacity Shila poured oil 4 times into a gallon. How much oil did she pour?
5. How many 250 ml glasses of squash can we make out of the squash bottle with the capacity of 1.5 litre?
6. If 15 glasses having capacity 180 ml can fill a jug then what will be the capacity of the jug?
7. A dairy fills milk in 500 ml packets. How many plastic packets will the dairy require if it sells 40,000 litres of milk?

Tip to the teacher: For farther multiplication and division exercise of capacity (l and ml related) design your own exercise similar to that of exercise?

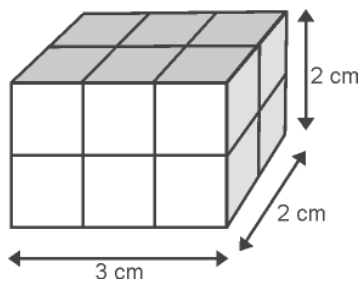
Lesson 10

VOLUME

Volume of a rectangular solid object:



In the picture a cube is shown with length, breadth and height 1 cm each. Its volume = 1 cube cm. This unit cube of 1 cube cm is the unit of measuring a rectangular solid figure.



To find out the volume of a rectangular solid object we count the number of unit cubes it contains. The rectangular solid object in the picture has length 3 cm, breadth 2 cm and height 2 cm. there are 6 unit cubes in the lower row 3 length wise and 2 breadth wise. Likewise there are 6 unit cubes on the upper row as well. The volume of each unit cube is 1 cube

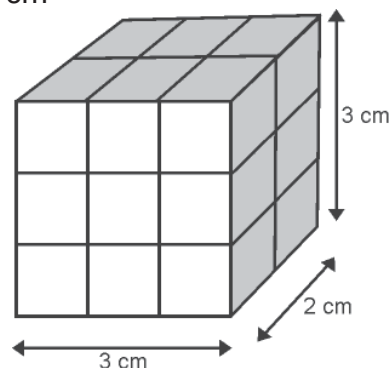
cm (1 cm^3). So the volume of the rectangular solid figure is 12 cube cm (12 cm^3).

Here, by multiplying length, breadth and height of the rectangular solid figure,

$$\begin{aligned} \text{Length} \times \text{breadth} \times \text{height} &= 3 \text{ cm} \times 2 \text{ cm} \times 2 \text{ cm} \\ &= 12 \text{ cm}^3 \end{aligned}$$

Now, if we add one more layer to the above picture we have to add 6 unit cubes. The volume of the rectangular figure is 18 cm^3 .

$$\begin{aligned} \text{Here, } l \times b \times h &= 3 \text{ cm} \times 2 \text{ cm} \times 3 \text{ cm} \\ &= 18 \text{ cm}^3 \end{aligned}$$



On the basis of the above discussion

The volume of a rectangular solid figure = length x breadth x height

$$\text{Or } V = l \times b \times h$$

Similarly, because the length, breadth and height of a cube is equal volume of cube (v) = (.....)³ or (length)³

$$\text{Or } V = l^3$$

Example 1

What is the volume of a rectangular figure which has length 4 cm, breadth 3 cm and height 2 cm?

$$\text{Here, length (l)} = 4 \text{ cm}$$

$$\text{Breadth (b)} = 3 \text{ cm}$$

$$\text{Height (h)} = 2 \text{ cm}$$

$$\text{Volume (V)} = ?$$

$$\text{So, } V = l \times b \times h$$

$$= 4 \text{ cm} \times 3 \text{ cm} \times 2 \text{ cm}$$

$$= 24 \text{ cm}^3$$

Example 2

What is the volume of a cube which 4 cm?

$$\text{Here, length of cube (l)} = 4 \text{ cm}$$

$$\text{Volume (V)} = ?$$

$$\text{So, volume of cube (V)} = l^3$$

$$= (4 \text{ cm})^3$$

$$= 4 \text{ cm} \times 4 \text{ cm} \times 4 \text{ cm}$$

$$\therefore V = 64 \text{ cm}^3$$

Example 3

What will be the volume of a soap with length 5 cm, breadth 4 cm and height 3 cm?

Here, length (l) = 5 cm

Breadth (b) = 4 cm

Height (h) = 3 cm

Volume (v) = ?

According to the formula,

$$\begin{aligned}v &= l \times b \times h \\&= 5 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm} \\&= 60 \text{ cm}^3\end{aligned}$$

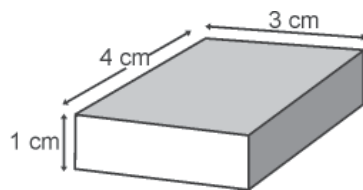
Volume of a soap (v) = 60 cm^3

$\therefore$

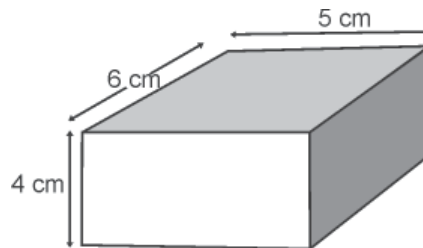
Exercise 10

(1) Find out the volume of each of the following rectangular figures.

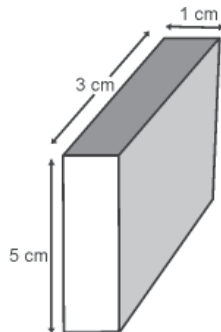
(a)



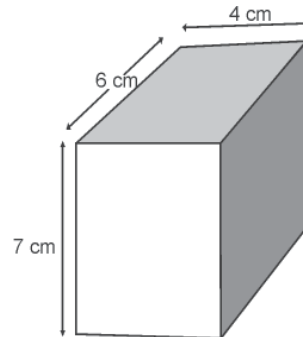
(b)



(c)

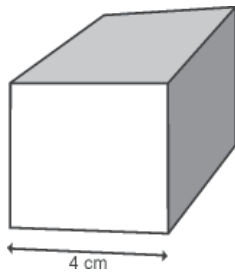


(d)

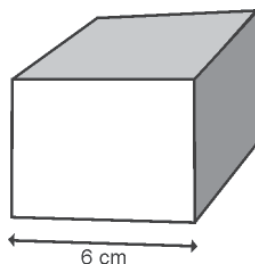


(2) Find out the volume of each of the following cubic figure.

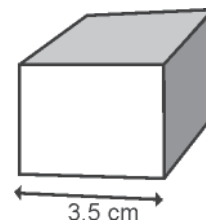
(a)



(b)



(c)



3. Find the volume of each of the following rectangular solid figures with given measurement:

- a. length = 4 cm, breadth = 3 cm, height = 2 cm
- b. length = 5 cm, breadth = 2 cm, height = 1 cm
- c. length = 3.5 cm, breadth = 2.2 cm, height = 4 cm
- d. length = 4.8 cm, breadth = 3.3 cm, height = 2.5 cm

4. A match box has length 13 cm, breadth 3 cm and height 3 cm. What will be its volume?

5. What will be the volume of a tooth paste with length 13 cm, breadth 3 cm and height 3 cm?

6. The cover of an ink-bottle has length 8 cm, breadth 4 cm and height 5 cm. What will be the volume of the cover? What will be the total volume of 18 such covers?

7. While making a table showing length, breadth and height of different rectangular solid objects Ram has put some boxes empty. The table is given below. Complete the empty boxes.

Measurement	a	b	c	d	e
Length(l)	3 cm	7 cm	6 cm	7 cm	5 cm
Breadth (b)	2 cm	5 cm	5 cm	6 cm	4 cm
Height(h)	4 cm	2 cm	4 cm	4 cm	4 cm
Volume (V)	?	?	?	?	?

Tips for teacher: While doing teaching learning activities on volume ask students to find out volume of rectangular solid objects found around the school and their home.

Lesson 9

CAPACITY

How much water does a glass contain?

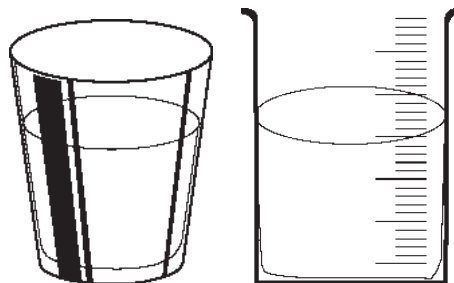
How much water fills up a water pot?

To answer such questions we have to measure the water in it. We use objects like cylinder or litre to measure liquid.

Units like litre and milliliters are used to measure water or other liquids. Capacity is the measurement of the amount of liquid in a container.

Litre and milliliters are the units used to measure capacity.

1 litre (l) = 1000 millilitre (ml)



Example 1

How many milliliters are there in 5 litres and 200ml.

Here, 5 l = 5000ml

= 5 × 5000 ml

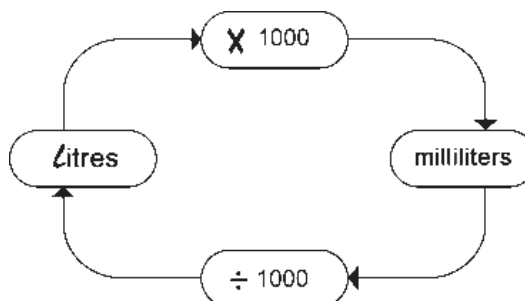
Now, 5l, 200ml = (5000 + 200) ml

= 5200ml

Example 2

Multiply:

$$\begin{array}{r} 1 \text{ L} \\ 2 \text{ 300} \\ \times 4 \\ \hline 8 \text{ 1200} \\ = 9 \text{ 1 200 L} \end{array}$$



Example 3

Solve : $15\cancel{1} 600\text{ m}\cancel{1} \div 5$

$$\begin{array}{r} 3\cancel{1} \quad 120\text{m}\cancel{1} \\ 5 \overline{) \quad 1 \quad \text{m}\cancel{1}} \\ \underline{15} \quad 600 \\ -15 \\ \hline \text{X} \quad 600 \\ \quad \underline{-5} \\ \quad \quad 10 \\ \quad \quad \underline{-10} \\ \quad \quad \quad 0 \end{array}$$

Example 4

Capacity of a glass is $250\text{ m}\cancel{1}$. Sujan drank 5 glasses of water at one time. How much water did he drink? Express in litres.

$$\begin{aligned} \text{Here, 5 glasses} &= 5 \times 250\text{ m}\cancel{1} \\ &= 1250\text{ m}\cancel{1} \\ &= 1250 \div 1000\cancel{1} \\ &= 1.250\text{ litres} \end{aligned}$$

Example 5

How many glasses with the capacity of $500\text{m}\cancel{1}$ are required to fill up a jug with a capacity of $2\cancel{1} 500\text{ m}\cancel{1}$?

Here,

$$\text{Capacity of jug} = 2\cancel{1} 500\text{ m}\cancel{1}$$

$$= (2 \times 1000 + 500 \text{ ml}) \text{ ml}$$

$$= 2500 \text{ ml}$$

Capacity of glass = 500ml

Therefore,

$$\begin{aligned} \text{Number of glasses required to fill up the bottle} &= \frac{2500}{500} \\ &= 5 \end{aligned}$$

∴ The jug will be filled up with 5 glasses.

Exercise 9

(1) Solve:

$$\begin{array}{r} \text{(a)} \quad \text{l} \quad \text{ml} \\ 3 \quad 400 \\ \hline \times 3 \end{array}$$

$$\begin{array}{r} \text{(b)} \quad \text{l} \quad \text{ml} \\ 15 \quad 250 \\ \hline \times 4 \end{array}$$

$$\begin{array}{r} \text{(c)} \quad \text{l} \quad \text{ml} \\ 21 \quad 250 \\ \hline \times 6 \end{array}$$

$$\begin{array}{r} \text{(d)} \quad \text{l} \quad \text{ml} \\ 16 \quad 750 \\ \hline \times 8 \end{array}$$

$$\text{(e)} 10\text{l}.200\text{ml} \div 3$$

$$\text{(f)} 15\text{l}.750\text{ml} \div 6$$

$$\text{(g)} 48\text{l}.500\text{ml} \div 5$$

$$\text{(h)} 28\text{l}.250\text{ml} \div 10$$

2. If a glass's capacity is 280 ml of water. How many glasses are required to fill a jug with the capacity of 4 l 200ml?
3. What capacity of kettle is required to serve 300ml of tea per person at once to 50 people?

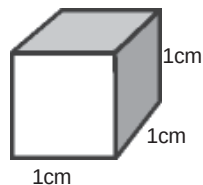
4. With a pot of 750 ml capacity Shila poured oil 4 times into a gallon. How much oil did she pour?
5. How many 250 ml glasses of squash can we make out of the squash bottle with the capacity of 1.5 litre?
6. If 15 glasses having capacity 180 ml can fill a jug then what will be the capacity of the jug?
7. A dairy fills milk in 500 ml packets. How many plastic packets will the dairy require if it sells 40,000 litres of milk?

Tip to the teacher: For farther multiplication and division exercise of capacity (l and ml related) design your own exercise similar to that of exercise?

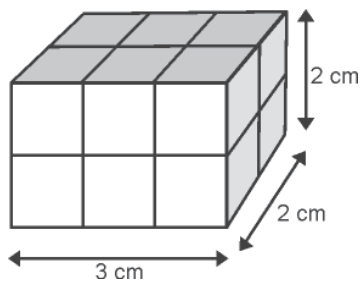
Lesson 10

VOLUME

Volume of a rectangular solid object:



In the picture a cube is shown with length, breadth and height 1 cm each. Its volume = 1 cube cm. This unit cube of 1 cube cm is the unit of measuring a rectangular solid figure.



To find out the volume of a rectangular solid object we count the number of unit cubes it contains. The rectangular solid object in the picture has length 3 cm, breadth 2 cm and height 2 cm. there are 6 unit cubes in the lower row 3 length wise and 2 breadth wise. Likewise there are 6 unit cubes on the upper row as well. The volume of each unit cube is 1 cube

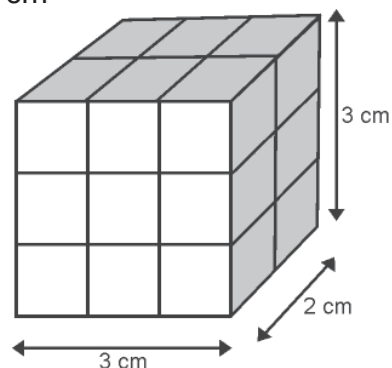
cm (1 cm³). So the volume of the rectangular solid figure is 12 cube cm (12 cm³).

Here, by multiplying length, breadth and height of the rectangular solid figure,

$$\begin{aligned}\text{Length} \times \text{breadth} \times \text{height} &= 3 \text{ cm} \times 2 \text{ cm} \times 2 \text{ cm} \\ &= 12 \text{ cm}^3\end{aligned}$$

Now, if we add one more layer to the above picture we have to add 6 unit cubes. The volume of the rectangular figure is 18 cm³.

$$\begin{aligned}\text{Here, } l \times b \times h &= 3 \text{ cm} \times 2 \text{ cm} \times 3 \text{ cm} \\ &= 18 \text{ cm}^3\end{aligned}$$



On the basis of the above discussion

The volume of a rectangular solid figure = length x breadth x height

$$\text{Or } V = l \times b \times h$$

Similarly, because the length, breadth and height of a cube is equal volume of cube (v) = (.....)³ or (length)³

$$\text{Or } V = l^3$$

Example 1

What is the volume of a rectangular figure which has length 4 cm, breadth 3 cm and height 2 cm?

Here, length (l) = 4 cm

Breadth (b) = 3 cm

Height (h) = 2 cm

Volume (V) = ?

$$\text{So, } V = l \times b \times h$$

$$= 4 \text{ cm} \times 3 \text{ cm} \times 2 \text{ cm}$$

$$= 24 \text{ cm}^3$$

Example 2

What is the volume of a cube which 4 cm?

Here, length of cube (l) = 4cm

Volume (V) = ?

$$\text{So, volume of cube (V) } = l^3$$

$$= (4\text{cm})^3$$

$$= 4\text{cm} \times 4\text{cm} \times 4\text{cm}$$

$$\therefore V = 64\text{cm}^3$$

Example 3

What will be the volume of a soap with length 5 cm, breadth 4 cm and height 3 cm?

Here, length (l) = 5 cm

Breadth (b) = 4 cm

Height (h) = 3 cm

Volume (v) = ?

According to the formula,

$$\begin{aligned} v &= l \times b \times h \\ &= 5 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm} \\ &= 60 \text{ cm}^3 \end{aligned}$$

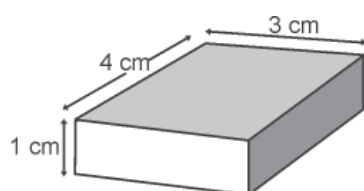
Volume of a soap (v) = 60 cm³

∴

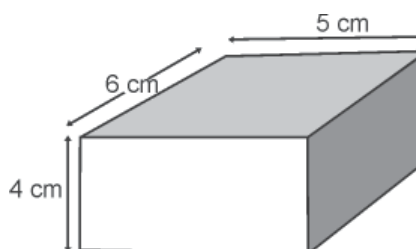
Exercise 10

(1) Find out the volume of each of the following rectangular figures.

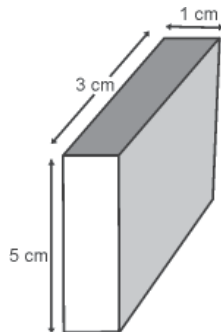
(a)



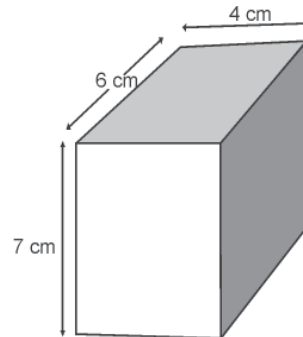
(b)



(c)

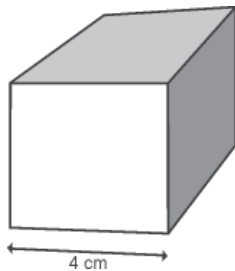


(d)

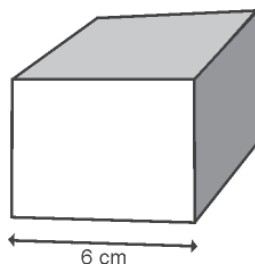


(2) Find out the volume of each of the following cubic figure.

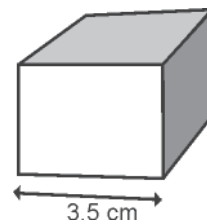
(a)



(b)



(c)



3. Find the volume of each of the following rectangular solid figures with given measurement:

- a. length = 4 cm, breadth = 3 cm, height = 2 cm
- b. length = 5 cm, breadth = 2 cm, height = 1 cm
- c. length = 3.5 cm, breadth = 2.2 cm, height = 4 cm
- d. length = 4.8 cm, breadth = 3.3 cm, height = 2.5 cm

4. A match box has length 13 cm, breadth 3 cm and height 3 cm. What will be its volume?

5. What will be the volume of a tooth paste with length 13 cm, breadth 3 cm and height 3 cm?

6. The cover of an ink-bottle has length 8 cm, breadth 4 cm and height 5 cm. What will be the volume of the cover? What will be the total volume of 18 such covers?
7. While making a table showing length, breadth and height of different rectangular solid objects Ram has put some boxes empty. The table is given below. Complete the empty boxes.

Measurement	a	b	c	d	e
Length(l)	3 cm	7 cm	6 cm	7 cm	5 cm
Breadth (b)	2 cm	5 cm	5 cm	6 cm	4 cm
Height(h)	4 cm	2 cm	4 cm	4 cm	4 cm
Volume (V)	?	?	?	?	?

Tips for teacher: While doing teaching learning activities on volume ask students to find out volume of rectangular solid objects found around the school and their home.

Lesson 11

WEIGHT

What will be the weight of " My Mathematics Class 5" ? Guess.

What is used to measure weight? What is the unit of weight? We use a weighting scale to measure weight? The weights used in the balance are of various weights like 100 gram, 200 gram, 500 gram. Kilogram etc.

Read the following facts about weight:

Weight of 1 kilogram = 2 weights of $\frac{1}{2}$ kilogram each.

= 5 weights of 200 gram

= 10 weights of 100 gram

= 1000 gram

100 kilogram = 1 quintal

Example 1

Convert 3.5 kg into gram.

Here, 1 kg is equal to 1000 gram.

Hence, 3.5 kg = $3.5 \times 1000 = 3500$ gram.

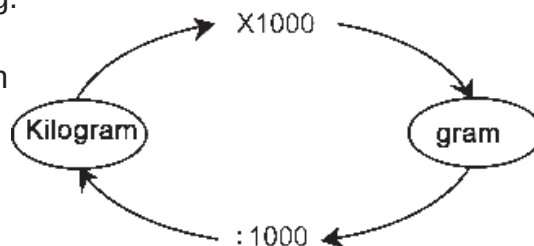
Example 2

Convert 75 gram into kg:

Here, 1000 gram is equal to 1 kg

Hence, 75 gram = $\frac{75}{1000}$ kg = 0.075 kg.

We can make the following conversion circle to convert gram and kilogram.



Example 3

The weight of a packet of tea is 1 kg 200 gm. What will be the total weight of 8 such packets?

Here, the weight of 8 packets of tea is eight times more than 1 packet of tea. By multiplying kg and gram separately,

$$\begin{array}{r} \text{kg} \quad \text{gram} \\ 1 \quad 250 \\ \times \quad 8 \\ \hline 8 \quad 2000 \\ \hline 10 \quad 000 \end{array} \longrightarrow \begin{array}{l} 2000 \text{ gram} = 2 \text{ kg} \\ = 10 \text{ gram} \end{array}$$

Lets do it another way, Weight of 8 packets of tea

$$\begin{array}{r} 1.25 \\ \times 8 \\ \hline 10.000\text{kg} \\ = 10\text{kg}. \end{array}$$

Example 4

What will be the weight of 1 packet of chocolate if the weight of 7 packets of chocolate is 5 kg 200 gram?

Here, to find out the weight of 1 packet of chocolate we have to divide 5 kg 250 gram into 7 equal divisions. Therefore,

$$\begin{array}{r} 0 \text{ kg} \quad 750 \text{ gram} \\ 7 \overline{) \text{ kg} \quad \text{gram}} \\ \underline{5 \quad 250} \\ 0 \quad 5250 \longrightarrow 5\text{kg.} = 5000 \text{ gram} \\ \underline{- 49} \\ 35 \\ \underline{- 35} \\ 00 \end{array}$$

Hence, the weight of 1 packet of chocolate = 0.750 kg or 750 gram.

Example 5

What will be the total weight of 2 packet biscuits each with 450 gram weight, 4 packet tea each with 125 gram weight and 3 butters each with $\frac{1}{2}$ kg weight ?

Here, weight of 2 packet biscuits = 2×450 gram
= 900 gram

Weight of 4 packet tea = 4×125 gram
= 500 gram

Weight of 3 butters = 3×500 gram
= 1500 gram

Hence, total weight = 900 gram + 500 gram + 1500 gram
= 2900 gram
= 2 kg 900 gram
= 2.9 kg

Exercise 11

(1) Convert into grams:

a. 5 kg

b. 12 kg

c. 16 kg

d. $\frac{1}{2}$ kg

e. $\frac{1}{4}$ kg

f. $6\frac{1}{2}$ kg

g. $7\frac{1}{5}$ kg

h. $5\frac{1}{4}$ kg.

i. 0.2 kg

j. 0.34 kg

k. 5.05 kg

l. 0.005 kg

(2) **Convert into Kilogram:**

- a. 3000 gram b. 5000 gram c. 2100 gram
d. 350 gram e. 250 gram f. 200 gram
g. 15 gram h. 90 gram i. 5 gram
j. 2 quintals k. 5 quintals

(3) **Multiply:**

(a) $\begin{array}{r} 2\text{kg } 250\text{gram} \\ \times 5 \\ \hline \end{array}$	(b) $\begin{array}{r} 6\text{kg } 720\text{gram} \\ \times 6 \\ \hline \end{array}$
(c) $\begin{array}{r} 5\text{kg } 650\text{gram} \\ \times 8 \\ \hline \end{array}$	(d) $\begin{array}{r} 8\text{ kg } 105\text{gram} \\ \times 9 \\ \hline \end{array}$
(e) $\begin{array}{r} 16\text{kg } 270\text{gram} \\ \times 5 \\ \hline \end{array}$	(f) $\begin{array}{r} 17\text{kg } 350\text{gram} \\ \times 7 \\ \hline \end{array}$

4. Divide:

- a. 500 g by 25 b. 750 g by 15
c. 3 kg by 650g by 6 d. 5 kg 460g by 4
e. 9 kg 300g by 6 f. 12 kg 420g by 9

5. There are 36 packets of powder in a box. The weight of one packet is 750 g and the weight of the box is 2.3 kg. what will be the total of the powder and the box?

6. If the weight of a tea-cup is 350 g,

- a. What will be the weight of 12 tea-cups?
b. How many tea-cups will make the weight of 1.05 kg?

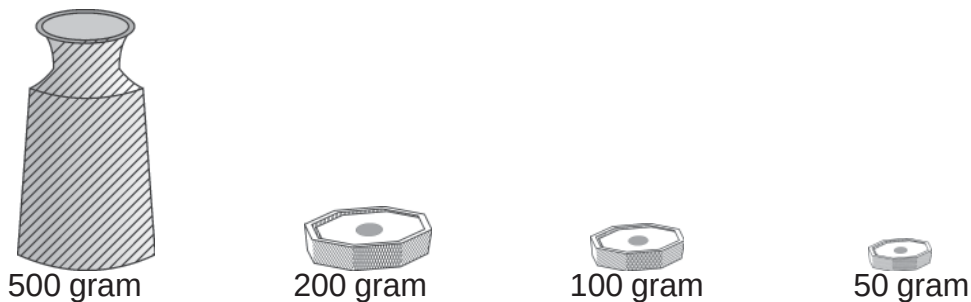
7. If the weight of 1 ink-pot is 270 g-

- a. What will be the weight of 14 ink-pots?
- b. How many ink-pots will make the weight of 5.4 kg?

8. 15.6 kg of beaten rice were distributed equally among 30 students. What is the weight of beaten rice each child got?

9. $3\frac{1}{4}$ kg of apples were distributed equally among 25 people. What is the weight of the apple each child got?

10. Using only one unit of the following weight how can a weighing scale take the weight of only 400 g at a time?



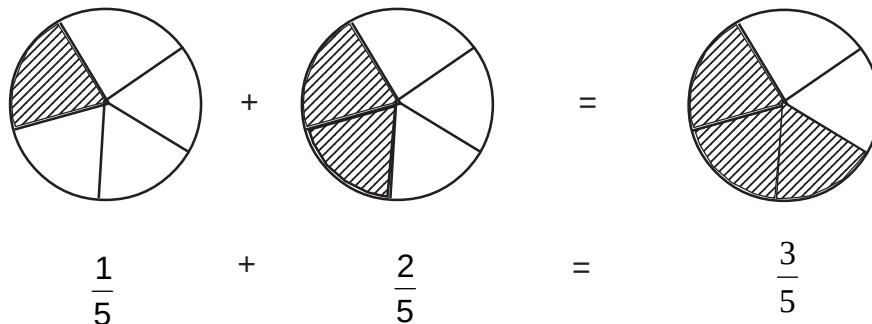
Tips for the teacher: While doing the teaching learning activities on weight ask students to guess the weight of the solid objects around them. Then tell them to take the weight of these objects and ask them to compare their guesses with the real weight.

Lesson 12

FRACTION AND DECIMAL

12.1 Additional of mixed numeral

Mother gave a bread to Shila. Shila divided the bread into 5 equal pieces and ate up 1 piece first. After some time she ate up 2 pieces. Thus, she ate up 3 pieces out of 5. Showing it in picture,



To express it mathematically,

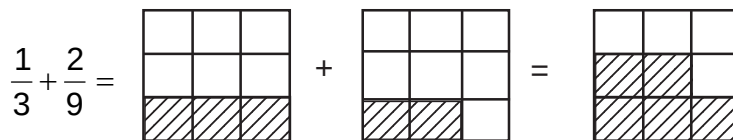
$$\begin{aligned} & \frac{1}{5} + \frac{2}{5} \\ &= \frac{1+2}{5} \\ &= \frac{3}{5} \end{aligned}$$

If the fraction to be added have same denominator we have to add only the numerators. The denominator remains the same.



Example 1

$$\begin{aligned}\frac{1}{3} + \frac{2}{9} &= \\&= \frac{1 \times 3}{3 \times 3} + \frac{2}{9} \\&= \frac{3}{9} + \frac{2}{9} \\&= \frac{3+2}{9} = \frac{5}{9}\end{aligned}$$



To make the same denominator of both the fractions we multiply both the numerator and denominator of 3 by 3,

$$\frac{1}{3} = \frac{1 \times 3}{3 \times 3} = \frac{3}{9}.$$

If the denominators to be added are different addition are made after converting them into same denominator.

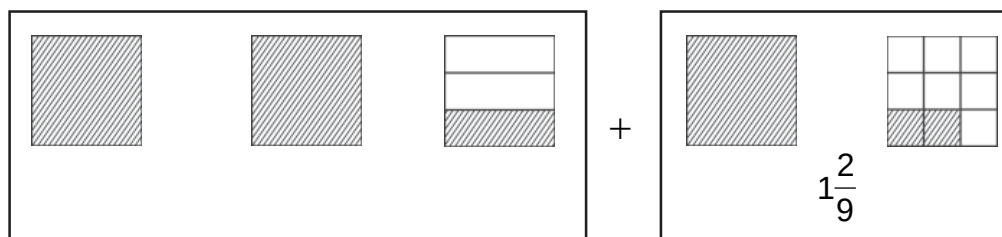
Example 2

Add:

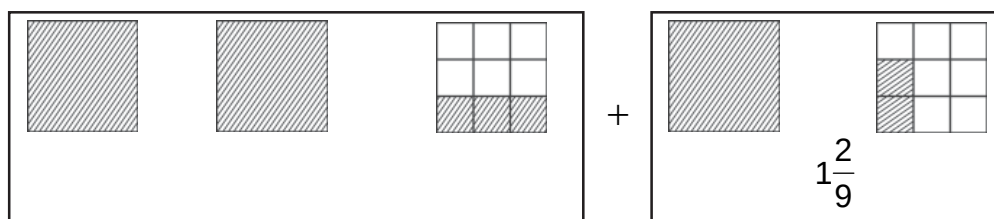
$$2\frac{1}{3} + 1\frac{2}{9}$$

Here, the numerals to be added are mixed numerals. Look at the following pictures

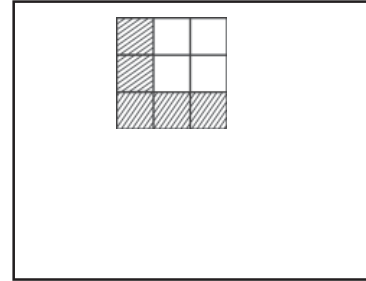
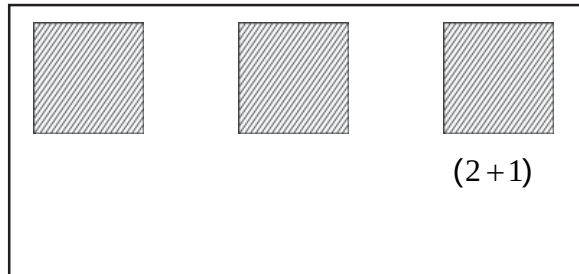
$$2\frac{3}{9}$$



Now, to make common denominator,



While adding whole and parts together -



$$\text{Therefore, } 2\frac{1}{3} + 1\frac{2}{9} = 2\frac{3}{9} + 1\frac{2}{9} = (2 + 1) + \left(\frac{3}{9} + \frac{2}{9}\right) = 3 + \frac{5}{9} = 3\frac{5}{9}$$

Example 3

Add:

$$5\frac{3}{8} + 3\frac{3}{4}$$

Here,

$$\left(\frac{3}{9} + \frac{2}{9}\right) = \frac{5}{9}$$

$$5\frac{3}{8} + 3\frac{3}{4}$$

$$= (5 + 3) + \left(\frac{3}{8} + \frac{3}{4}\right)$$

$$= 8 + \left(\frac{3}{8} + \frac{3 \times 2}{4 \times 2}\right)$$

$$= 8 + \left(\frac{3}{8} + \frac{6}{8}\right)$$

$$= 8 + \frac{9}{8} \longrightarrow \left[\frac{9}{8}\right] \text{ is improper fraction.}$$

$$= 8 + 1\frac{1}{8} = (8 + 1) + \frac{1}{8} = 9\frac{1}{8}$$

Exercise 12.1

Solve:

$$1 \quad 2\frac{1}{2} + 1\frac{1}{2}$$

$$2 \quad 3\frac{1}{2} + 4\frac{1}{4}$$

$$3 \quad 3\frac{1}{4} + 2\frac{3}{8}$$

$$4 \quad 4\frac{1}{3} + 5\frac{2}{9}$$

$$5 \quad 4\frac{2}{5} + 3\frac{3}{10}$$

$$6 \quad 3\frac{1}{4} + 2\frac{5}{12}$$

$$7 \quad 3\frac{3}{4} + 1\frac{1}{2}$$

$$8 \quad 6\frac{2}{5} + 5\frac{4}{15}$$

$$9 \quad 8\frac{5}{6} + 1\frac{1}{12}$$

$$10 \quad 3\frac{1}{4} + 5\frac{1}{12}$$

$$11 \quad 7\frac{1}{3} + 2\frac{5}{6}$$

$$12 \quad 4\frac{2}{7} + 1\frac{11}{14}$$

$$13 \quad 10\frac{2}{3} + 7\frac{1}{6}$$

$$14 \quad 3\frac{5}{6} + 2\frac{5}{12}$$

$$15 \quad 9\frac{3}{11} + 6\frac{21}{22}$$

12.2 Subtraction of Mixed Numbers

Like addition of mixed numbers, subtraction of mixed numbers can be done by subtracting whole number from whole number and fraction number from fraction numbers.

Example 1

Subtract:

$$6\frac{1}{3} - 3\frac{2}{9}$$

Here,

$$\begin{aligned} 6\frac{1}{3} - 3\frac{2}{9} &= (6 - 3) + \left(\frac{1}{3} - \frac{2}{9}\right) \longrightarrow \text{(subtracting single number from single number \& fraction from fraction).} \\ &= 3 + \left(\frac{1 \times 3}{3 \times 3} - \frac{2}{9}\right) = 3 + \left(\frac{3}{9} - \frac{2}{9}\right) \\ &= 3 + \left(\frac{3 - 2}{9}\right) = 3 + \frac{1}{9} = 3\frac{1}{9} \end{aligned}$$

Example 2

Simplify:

$$10\frac{3}{4} - 5\frac{7}{8}$$

Here,

$$\begin{aligned} 10\frac{3}{4} - 5\frac{7}{8} &= (10 - 5) + \left(\frac{3}{4} - \frac{7}{8}\right) = 5 + \left(\frac{3 \times 2}{4 \times 2} - \frac{7}{8}\right) \\ &= 5 + \left(\frac{6}{8} - \frac{7}{8}\right) = 4 + \left(\frac{8}{8} + \frac{6}{8} - \frac{7}{8}\right) \longrightarrow \text{(since 7 is not subtracted from 6, whole number 1 = } \frac{8}{8} \text{ is borrowed from 5).} \\ &= 4 + \left(\frac{8 + 6 - 7}{8}\right) = 4 + \frac{7}{8} = 4\frac{7}{8} \end{aligned}$$

Example 3

$\frac{1}{3}$ of a part of garden has orange trees and $\frac{1}{6}$ part of it has mango trees. If pineapple is planted in the remaining area what parts of the garden does it cover?

Here, the parts covered by orange and mangoes

$$\begin{aligned} &= \frac{1}{3} + \frac{1}{6} \\ &= \frac{1 \times 2}{3 \times 2} + \frac{1}{6} \\ &= \frac{2+1}{6} \\ &= \frac{3}{6} = \frac{1}{2} \text{ parts} \end{aligned}$$

Therefore, the remaining part

$$\begin{aligned} &= 1 - \frac{1}{2} \\ &= \frac{2}{2} - \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

Changing the whole number into fraction 2.

Or

$$\begin{aligned} &\longrightarrow 1 - \left(\frac{1}{2} + \frac{1}{6} \right) \\ &= 1 - \left(\frac{2+1}{6} \right) \\ &= 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

Hence, the parts covered by pineapple = $\frac{1}{2}$

Exercise 12.2

(1) **Subtract:**

(a) $7\frac{5}{8} - 3\frac{1}{4}$

(b) $4\frac{1}{5} - 2\frac{3}{10}$

(c) $15 - 13\frac{1}{2}$

(d) $5\frac{3}{7} - 3\frac{5}{14}$

(e) $8\frac{13}{14} - 7\frac{1}{2}$

(f) $10\frac{4}{5} - 3\frac{3}{10}$

(2) **Simplify:**

(a) $2\frac{4}{7} - 1\frac{3}{14}$

(b) $13\frac{1}{4} - 9\frac{11}{12}$

(c) $25\frac{4}{5} - 20\frac{11}{15}$

(d) $87\frac{3}{13} - 37\frac{5}{26}$

(e) $18\frac{7}{8} - 6\frac{33}{40}$

(f) $12\frac{1}{13} - 7\frac{4}{39}$

(g) $1\frac{3}{4} - \frac{5}{6} + 2\frac{1}{2}$

(h) $5 - 1\frac{1}{2} - \frac{5}{8}$

(i) $3\frac{1}{4} - 2\frac{3}{4} + 2\frac{1}{2}$

3. If a student does $\frac{1}{2}$ fraction of work and another student does $\frac{1}{4}$ fraction of it what fraction of the work remains to be done?
4. Two parties got $\frac{1}{2}$ and $\frac{1}{3}$ fraction of the total vote cast. The rest of the votes were invalid. What fraction of the votes were invalid?
5. Out of the total money in the purse Ram took $\frac{1}{2}$ and Shyam took $\frac{3}{10}$ fractions. If Mahesh took the rest what fraction of the money did he take?
6. Of all the seats in a cinema hall $\frac{3}{7}$ fraction is for first class and $\frac{5}{14}$ fraction is for second class. What fraction of the seats are there for third class?

12.3 Multiplication of Fraction

a. Multiplication of a fraction by whole number.

While multiplying whole number multiplication means continuous adding.

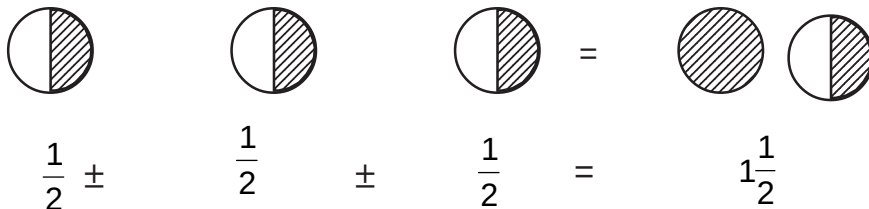
3×4 means 3 times 4.

Hence, $3 \times 4 = 4 + 4 + 4 = 12$.

Hence, $3 \times \frac{1}{2}$ means 3 times $\frac{1}{2}$.

Hence, $3 \times \frac{1}{2} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} = 1\frac{1}{2}$

To show it in picture –



$$\text{Or, } 3 \times \frac{1}{2} = \frac{3 \times 1}{2} = \frac{3}{2} = 1\frac{1}{2}$$

In multiplication of whole number and fraction, the numerator of the fraction will be multiplied by the whole number and the denominator will remain the same. If the result is the improper fraction it should be expressed in mixed number.

Example 1

Multiply:

$$5 \times \frac{7}{12}$$

Here,

$$\begin{aligned} 5 \times \frac{7}{12} &= \frac{5 \times 7}{12} \\ &= \frac{35}{12} = 2\frac{11}{12} \end{aligned}$$

Note: converting $\frac{35}{12}$ into mixed numeral numerator is divided by denominator. As for example:

$$\begin{array}{r} 2 \\ 12 \overline{)35} \\ \underline{24} \\ 11 \end{array}$$

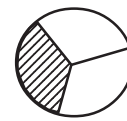
Hence, $\frac{35}{12} = 2\frac{11}{12}$

b. Multiplication of fraction and fraction

$\frac{1}{2} \times \frac{1}{3}$ means $\frac{1}{2}$ times $\frac{1}{3}$.

It means half of $\frac{1}{3}$. In figure $\frac{1}{3}$ when $\frac{1}{2}$ each part divided by $\frac{1}{2}$ shows $\frac{1}{6}$.

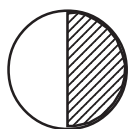
While thinking in other way we can say one third of $\frac{1}{2}$.



Dividing $\frac{1}{2}$ of each part.



$$\begin{aligned} \text{Coloured Part} &= \frac{1}{3} \text{ of } \frac{1}{2} \\ &= \frac{1}{6} \end{aligned}$$



Dividing one third of each part.



$$\begin{aligned} \text{Coloured part} &= \frac{1}{2} \times \frac{1}{3} \\ &= \frac{1}{6} \end{aligned}$$

Think in this way:

$$\frac{1}{2} \times \frac{1}{3} = \frac{1 \times 1}{2 \times 3} = \frac{1}{6}$$

When multiplying fractions each numerator should be multiplied by numerator and denominator by denominator

Example 2

Multiply:

$$\frac{3}{4} \times \frac{5}{7}$$

$$\frac{3 \times 5}{4 \times 7} = \frac{15}{28}$$

Exercise 12.3

(1) **Multiply:**

(a) $7 \times \frac{1}{3}$

(b) $8 \times \frac{1}{5}$

(c) $4 \times \frac{2}{9}$

(d) $9 \times \frac{9}{10}$

(e) $12 \times \frac{1}{5}$

(f) $15 \times \frac{11}{12}$

(g) $6 \times \frac{1}{8}$

(h) $4 \times \frac{1}{12}$

(i) $6 \times \frac{5}{12}$

(j) $\frac{3}{5} \times \frac{2}{5}$

(k) $\frac{6}{7} \times \frac{1}{5}$

(l) $\frac{2}{7} \times \frac{3}{5}$

(2) **Simplify:**

(a) $\frac{3}{5} \times \frac{2}{5}$

(b) $\frac{6}{7} \times \frac{1}{5}$

(c) $\frac{2}{7} \times \frac{3}{5}$

(d) $\frac{3}{10} \times \frac{5}{12}$

(e) $\frac{2}{5} \times \frac{9}{10}$

(f) $\frac{4}{11} \times \frac{7}{9}$

$$(g) \quad \frac{5}{11} \times \frac{7}{13}$$

$$(h) \quad \frac{6}{7} \times \frac{8}{9}$$

$$(i) \quad \frac{11}{12} \times \frac{1}{7}$$

(3) Draw figure that shows multiplication from below:

$$(a) \quad 5 \times \frac{1}{7}$$

$$(b) \quad 3 \times \frac{1}{7}$$

$$(c) \quad 4 \times \frac{2}{3}$$

12.4 Decimal

Fraction having denominator 10 or multiplier of 10 is called decimal.

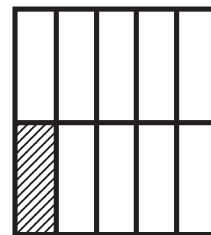
In the figure coloured part shows one tenth $\frac{1}{10}$.

We can write, $\frac{1}{10} = 0.1$

Similarly one hundredth can be

written as- $\frac{1}{100} = 0.01$, $\frac{1}{100} = 0.01$

One thousandth = $\frac{1}{1000} = 0.001$



$$\frac{1}{10} = 0.1$$

While converting fraction into decimal, denominator should be converted into 10 or multiple of 10 or numerator should be divided by denominator.

Example 1

Convert fraction into decimal:

$$\frac{1}{2} = \frac{1 \times 5}{2 \times 5} = \frac{5}{10} = 0.5$$

Example 2

Convert fraction into decimal:

$$\begin{aligned} 3\frac{1}{4} &= 3 + \frac{1}{4} \\ &= 3 + \frac{1 \times 25}{4 \times 25} = 3 + \frac{25}{100} \\ &= 3.25 \end{aligned}$$

Look at the place value table given below of decimal system. 13.204 has been shown in place value table.

Ten	One	Tenth	Hundredth	thousandth
10	1	$\frac{1}{10}$	$\frac{1}{100}$	$\frac{1}{1000}$
(10)	(1)	(0.1)	(0.01)	(0.001)
1	3	2	0	4

While converting decimal into fraction whole number should be kept as it is and its fractional form should be made by looking at what type of decimal it is tenth, hundredth or thousandth.

Example 3

Add:

$$\begin{array}{r} 16.10 \\ + 31.32 \\ \hline 47.42 \end{array}$$

Example 4

Subtract:

$$\begin{array}{r} 21.50 \\ - 16.59 \\ \hline 4.91 \end{array} \quad \longrightarrow \quad \text{Add 0 after decimal to make equal status with others.}$$

Exercise 12.4

(1) Convert the fractions given below to decimals:

- (a) $\frac{2}{10}$ (b) $\frac{3}{10}$ (c) $\frac{7}{10}$ (d) $\frac{1}{4}$ (e) $1\frac{1}{2}$
- (f) $21\frac{3}{10}$ (g) $13\frac{1}{3}$ (h) $15\frac{1}{5}$

(2) Convert the decimals given below into fraction:

- (a) 0.12 (b) 0.08 (c) 3.06 (d) 12.05
- (e) 0.25 (f) 0.009 (g) 17.012 (h) 0.005

(3) Make place value table of the numbers given below:

- (a) 1.35 (b) 13.01 (c) 15.12 (d) 0.05
- (e) 6.48 (f) 10.73 (g) 0.123 (h) 15.01

(4) Simplify:

- (a) $5.01 + 3.25$ (b) $6.07 + 3.2$ (c) $0.69 + 1.28$
- (d) $73.68 - 9.07$ (e) $15.04 - 11.06$ (f) $12 - 9.37$
- (g) $0.06 - 4.27 + 5.38$ (h) $4.5 - 8.25 + 5.07$

(5) How much more will be the sum of 3.91 and 6.04 than 2.46?

(6) From a ribbon of 50 cm long, if 16.80 cm and 14.25 cm are off taken how many cm is left.

(7) If Rs. 50 is given after buying a pen for Rs. 11.65 and dozen of copies for Rs. 36.25 . how much will be returned back?

(8) Distance from Pokhara to Kathmandu is 200km. If 103.675km is travelled through bus and 73.025km through taxi how much distance is left to travel?

12.5 Multiplication of decimal by 10, 100 and 1000.

$\frac{2}{1000} \times \frac{10}{1} = \frac{20}{1000}$ Two thousandth when multiplied by 10 becomes 20 thousandth. Saying 20 parts in thousand and 2 parts in hundred are same.

Therefore, $\frac{2}{1000} \times \frac{10}{1} = \frac{20}{1000}$ Or $0.002 \times 10 = 0.02$

Therefore, $\frac{2}{1000} \times \frac{100}{1} = \frac{200}{1000}$, Or $0.002 \times 100 = 0.2$

and $\frac{2}{1000} \times \frac{1000}{1} = \frac{2000}{1000}$, Or, $0.002 \times 1000 = 2$

Therefore, when decimal is multiplied by 10 decimal sign shifts one place right or gets removed. Similarly when decimal is multiplied by 100, 100 decimal sign shifts 2, 3 places right or gets removed respectively.

This process has been shown clearly in the table below:

Decimal	Multiplying number	What to do while multiplying?
0.001	10	$0.001 \times 10 = \overset{\curvearrowright}{0.0001} = 0.01$ shift decimal sign to one place right
0.001	100	$0.001 \times 100 = \overset{\curvearrowright}{0.001} = 0.1$ shift decimal sign to two places right
0.001	1000	$0.001 \times 1000 = \overset{\curvearrowright}{0.001} = 1.0$ Shift decimal sign three steps right.

Example 1

Multiply 10.3045 by 10, 100 and 1000 separately:

Here,

$$10.3045 \times 10 = 10.\overset{\curvearrowright}{3045} = 103.045$$

$$10.3045 \times 100 = 10.\overset{\curvearrowright}{3045} = 1030.45$$

$$10.3045 \times 1000 = 10.\overset{\curvearrowright}{3045} = 10304.5$$

Exercise 12.5

Multiply each numbers below by 10, 100 and 1000.

(1) 0.002 (2) 0.013 (3) 0.137 (4) 1.005

(5) 2.679 (6) 10.805 (7) 1.2 (8) 13.5

12.6 Multiplication of decimal by whole number.

See example about multiplication below

Example 1

(a) $3 \times 5 = 15$

(b) $3 \times 10 \times 5 = 30 \times 5 = 150$ or $3 \times 5 \times 10 = 3 \times 50 = 150$

$3 \times 5 = 15$ when either 3 or 5 is multiplied by 10 then product increases 10 times.

Example 2

$$3 \times 100 \times 5 = 300 \times 5 = 1500$$

$$\text{or } 3 \times 5 \times 100 = 3 \times 500 = 1500$$

$3 \times 5 = 15$ or $5 \times 3 = 15$, either 3 or 5 when multiplied by 100 the product increases by 100 times.

Based on these examples and exercise 12.5 multiplications on decimals can be done.

Example 3

Multiply:

$$0.3 \times 5$$

Here,

$\begin{array}{r} 0.3 \\ \times 5 \\ \hline 1.5 \end{array}$	$\begin{array}{r} 3 \\ \times 5 \\ \hline 15 \end{array}$	$\begin{array}{l} 0.3 \times 10 = 3, \text{ when} \\ \text{multiplied by } 5, 3 \times 5 \\ \text{becomes } 10 \text{ times } 0.3 \times 5. \\ \text{hence } 0.3 \times 5 = 3 \times 5 \div 10 \end{array}$
$\div 10$		

Other method

$$\begin{aligned} 0.3 \times 5 &= \frac{3}{10} \times 5 \\ &= \frac{15}{10} = 1.5 \end{aligned}$$

Example 4

Multiply:

$$0.03 \times 5$$

$\begin{array}{r} 0.03 \\ \times 5 \\ \hline 0.15 \end{array}$	$\begin{array}{r} 3 \\ \times 5 \\ \hline 15 \end{array}$	$\begin{array}{l} 0.03 \times 100 = 3 \text{ when multiplied} \\ \text{By } 5 \text{ we get } 15, \text{ which is hundredth} \\ \text{Times greater than } 0.03 \times 5. \\ \text{hence to get the result } 15 \\ \text{should be divided by } 100. \end{array}$
$\div 100$		

Other method

$$\begin{aligned} 0.03 \times 5 &= \frac{3}{100} \times 5 \\ &= \frac{15}{100} = 0.15 \end{aligned}$$

While calculating by both methods we got same result. What did you learn from this? What is similar between decimal sign in multiplier and decimal sign in product?

Therefore,



While multiplying decimals we use same process used in natural numbers and in answers we make equal positioning of decimals

Example 9

Multiply:

$$3.57 \times 12$$

Here,

$$\begin{array}{r} 3.57 \\ \times 12 \\ \hline 714 \\ 357 \\ \hline 42.84 \end{array}$$

This method is like basic multiplication process. In 3.57 there are two numbers after decimal sign so in product decimal sign must be placed before two numbers.

Exercise 12.6

(1) **Multiply:**

- | | | | |
|----------------------|-----------------------|------------------------|------------------------|
| (a) 0.4×2 | (b) 0.5×3 | (c) 0.3×7 | (d) 0.2×2 |
| (e) 3×0.8 | (f) 5×0.6 | (g) 8×0.4 | (h) 9×0.9 |
| (i) 0.05×5 | (j) 0.03×4 | (k) 0.07×6 | (l) 0.08×9 |
| (m) 6×0.12 | (n) 7×0.45 | (o) 9×0.99 | (p) 7.45×6 |
| (q) 15.02×3 | (r) 14×0.235 | (s) 21.096×12 | (t) 17.651×13 |

(2) **Multiply:**

- | | | |
|---------------------------------|--------------------------------|--------------------------|
| (a) 32.4×0.3 | (b) 7.24×0.5 | (c) 17.32×2.3 |
| (d) 42.07×3.4 | (e) 33.33×0.9 | (f) 56.6×3.23 |
| (g) 76.5×3.02 | (h) 0.05×0.02 | (i) 33.07×14.04 |
| (j) $2.5 \times 2.5 \times 2.5$ | (k) $7 \times 0.7 \times 0.07$ | |

- (3) **Find the total cost of every situation given below:**
- 12 oranges at the rate of Rs 2.25 per piece.
 - 15 copies at the rate of Rs 4.75 per piece.
 - 32 letter covers at the rate of Rs 0.75 per piece.
 - 35 pencils at the rate of Rs 1.05 per piece.
- 4) **With first payment of Rs 500, and installments of Rs 205.75 per month to cover total cost in 12 months, how much would be the total cost of sewing machine?**
- 5) **For a vrectangular object volume = length x breadth x height. Using this forlula find the volume of the rectangular objects given below:**
- Length = 1.2 cm, breadth = 0.8 cm and height = 1.1 cm.
 - Length = 4.5 cm, breadth = 2.2 cm and height = 1.8 cm.

12.7 Rounding off decimals

Among 8 persons Rs 9 is distributed eqaully then how much will each receive?
Here, 9 is divided by 8

$$\begin{array}{r} 1 \\ 8 \overline{)9} \\ - 8 \\ \hline 1 \end{array}$$

Therefore, each will receive Rs $1\frac{1}{8} =$

Representing in decimal, $\frac{1}{8} = 0.125$

Therefore each will receive Rs 1.125. this means that each will recieve Rs 1 and out of 1000 parts of Rs 1 they get 125 part.

But in daily practice this is not possible.

Means Rs 1 cannot be divided equally in 1000 parts. So what we should do? In this type of situation we divide to each nearly Rs 1.13 or Rs 1 and 13 paise. Here making Rs 1.125 = Rs 1.13 means in Rs 1.125 we are placing 0 in thousandth or rounding off in hundredth. While rounding off like this the place where we place 0 if there is number equal to 5 or greater than 5 then we zero that number and add one number to its left. If the rounding off place has number less than 5 then we make that number zero and write the left side number as it is.

For Example, 5.20735

= 502074, rounding off at 4th place of decimal

= 5.207, rounding off at 3rd place of decimal

= 5.21, rounding off at 2nd place of decimal

= 5.2, rounding off at 1st place of decimal

Exercise 12.7

1. **Round off at 1st place of decimal:**

- (a) 8.53 (b) 2.67 (c) 4.07 (d) 13.51

2. **Round off at 2nd place of decimal:**

- (a) 4.821 (b) 3.456 (c) 0.493 (d) 3.008

3. **Round off at 3rd place of decimal :**

- (a) 3.4156 (b) 4.0051 (c) 13.5305 (d) 15.6708

4. **Round off the given numbers according to the places given in the bracket:**

- | | |
|-----------------|-----------------|
| (a) 5.635 (2) | (b) 1.8918 (3) |
| (c) 15.3445 (2) | (d) 3.0729 (1) |
| (e) 0.00581 (2) | (f) 3.14159 (3) |

Lesson 13

PERCENTAGE

13.1 Meaning of percentage

Dhruva scored the following in the exam:

Subject	Nepali	English	Math	Science
Score	16	19	34	45
Full mark	20	25	50	100

In which subject did Dhruva perform best?

In the table the score of science is 45 and that of Nepali is 16. From this can we say that Dhruva did best in science and worst in Nepali?

Here, writing it in fraction the basis of the score and full mark,

$$\text{Nepali, } \frac{16}{20}$$

$$\text{English, } \frac{19}{25}$$

$$\text{Math, } \frac{34}{50}$$

$$\text{Science, } \frac{45}{100}$$

Which is the largest fraction? How do we know it? Converting them into fractions with the same denominator,

$$\text{Nepali, } \frac{16}{20} = \frac{16 \times 5}{20 \times 5} = \frac{80}{100}$$

$$\text{English, } \frac{19}{25} = \frac{19 \times 4}{25 \times 4} = \frac{76}{100}$$

$$\text{Math, } \frac{34}{50} = \frac{34 \times 2}{50 \times 2} = \frac{68}{100}$$

$$\text{Science, } \frac{45}{100}$$

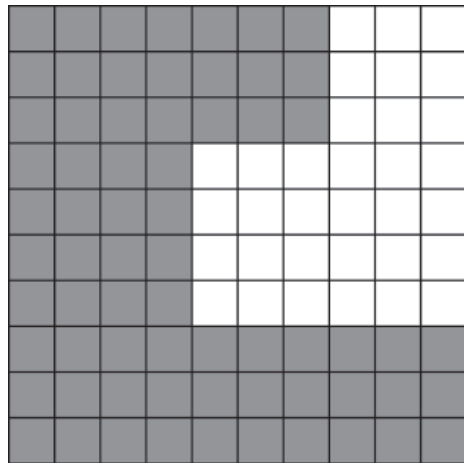
From this we can say that Dhruva did best in Nepali and worst in Science.

Here the denominator of each fraction is 100. Therefore, the numerator of each fraction denotes the percentage of the score. The sign % denotes percentage.

Here, Dhruba has score 80% in Nepali, 76% in English, 68% in maths and 45% in Science.

If the denominator of a fraction is 100 the numerator denotes the percentage which shows how much out of each hundred. In the adjoining picture 67 out of 100 percent shaded.

It is written $\frac{67}{100} = 67\%$ and is read as 67 percentage. Percentage helps us to compare two or more quantities.



13.2 Converting fractions into percentage:

Convert $\frac{3}{4}$ into percentage:

Method -1

$$\frac{3}{4} = \frac{3 \times 25}{4 \times 25} \quad (\text{suppose the denominator is 100})$$

$$= \frac{75}{100} = 75\%$$

Method - 2

$\frac{3}{4}$ means 3 out of 4 parts.

It is $\frac{3}{4}$ of 1 part.

$$\begin{aligned} \text{Hence, it is out of 100 parts } & \frac{3}{4} \times \frac{25}{1} \times 100 \\ & = 3 \times 25\% = 75\% \end{aligned}$$

While converting a fraction into percentage the given fraction should be multiplied by 100.

13.3 Converting percentage into fraction:

Example:

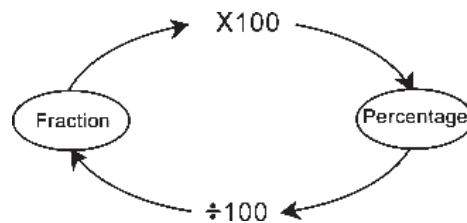
Convert 75% into fraction.

Here, 75% means 75 out of 100 parts.

$$\text{Hence, } 75\% = \frac{75}{100} = \frac{3}{4}$$

To convert percentage into fraction given number of percentage should be divided by 100.

Based on above discussion method of converting Fraction and percentage into each other is shown in the circle on the right.



Exercise 13.1

(1) Write in percentage:

(a) $\frac{25}{100}$

(b) $\frac{27}{100}$

(c) $\frac{33}{100}$

(d) $\frac{45}{100}$

(e) $\frac{63}{100}$

(f) $\frac{125}{100}$

(2) Convert into percentage (use both methods):

(a) $\frac{1}{2}$

(b) $\frac{3}{4}$

(c) $\frac{2}{5}$

(d) $\frac{7}{10}$

(e) $\frac{3}{5}$

(f) $\frac{17}{20}$

(g) $\frac{22}{25}$

(h) $\frac{33}{50}$

(i) $\frac{5}{2}$

(j) $\frac{7}{5}$

(3) Convert percentage into fraction:

(a) 15%

(b) 20%

(c) 25%

(d) 35%

(e) 50%

(f) 85%

(g) 48%

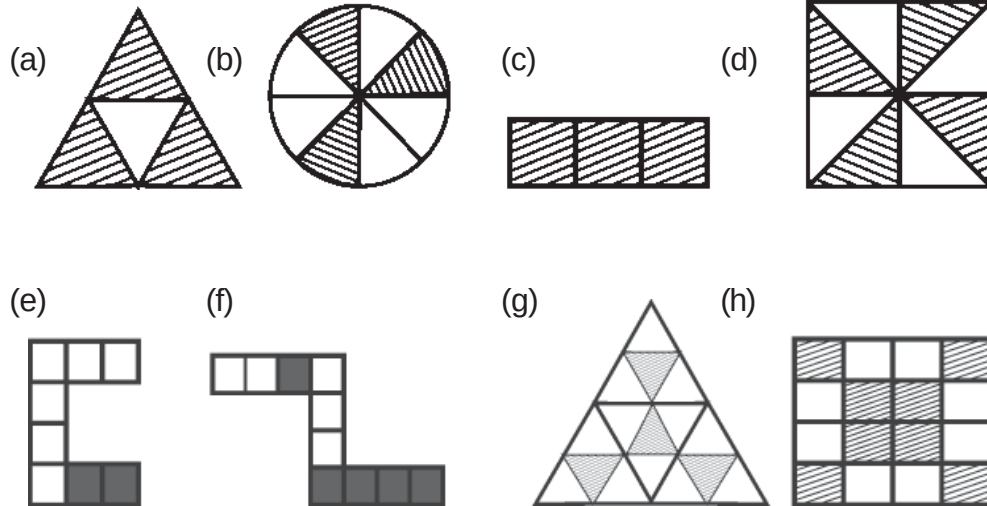
(h) 16%

(i) $12\frac{1}{2}\%$

(j) 115%

(k) $33\frac{1}{2}\%$

- (4) First write the shaded parts into fractions then change them into percentage:



- (5) Express the non shaded parts of each of the pictures in question 4 in percentage.
- (6) What will be the total of the addition of the non shaded percentage and shaded percentage of each of the picture in question number 4?

13.4 Use of percentage

We can use percentage to solve different practical problem of our daily life. Look at the following example:

Example 1

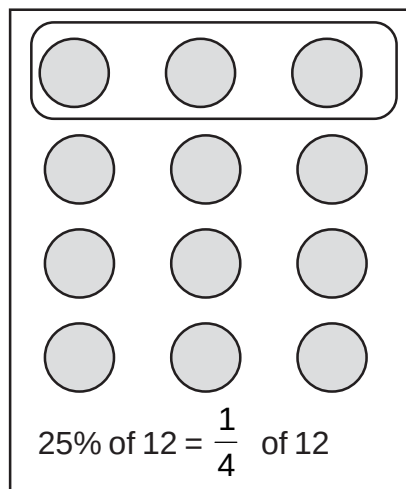
How much is the 25% of Rs 12?

Here, 25% of Rs 12/-

$$= \text{Rs } 12 \times \frac{25}{100} = \text{Rs } 12 \times \frac{1}{4}$$

$$= \text{Rs } 12 \times \frac{1}{4} \text{ (of means multiplication in fraction)}$$

$$= \text{Rs } 3$$



Example 2

There are 400 students in a school. Among them 35% are girls. What is the number of girls and boys?

Here, number of girls = 35% of 400

$$= 400 \text{ of } \frac{35}{100}$$

$$= 400 \times \frac{35}{100} = 400 \times \frac{7}{20}$$

$$= 140 \text{ persons}$$

$$\text{and boys \%} = (100 - 35)\% = 65\%$$

Hence, number of boys = 400 of 65%

$$= 400 \times \frac{65}{100} = 400 \times \frac{13}{20}$$

$$= 260 \text{ persons}$$

Can you find out another method of finding out boy's number?

Exercise 13.2

- (1) Find out value.
- | | |
|-----------------------------|-----------------------------------|
| a. 12% of 50 | b. 15% of 20 |
| c. 35% of 60 | d. 60% of Rs 5/- |
| e. 35% of 150 | f. 25% of 1 kg |
| g. $6\frac{2}{3}\%$ of 30km | h. $12\frac{1}{2}\%$ of 16 litres |
| i. 100ml of 200% | j. 150% of Rs 30/- |
- (2) **If there are 40 students in a class among which 15 are girls. What is the percentage of the number of boys and girls?**
- (3) **There are 30 eggs in a crate. If 5 of them are broken -**
- | |
|------------------------------------|
| a. How many percent is broken? |
| b. How many percent is not broken? |
- (4) **If Hari scored 45 out of 60 in Mathematics, how many percent did he score?**
- (5) **A sweater costs Rs 500/-. If 20% is discounted how many rupees if discounted?**

Lesson 14

UNITARY METHOD & SIMPLE INTEREST

14.1 Unitary method

On the following table the number of pencils and their costs are given. Look at the table and discuss the questions given below:

Pencils					
Cost	5	10	15	20	25

- Has the cost changed as the no. of pencils change?
- Can the cost of the pencil deduct the cost of required no. of pencils? How?
- Can we decide the cost of 1 pencil from the cost of some pencils? How?

According to the above table, to deduct the cost of some pencils from the cost of 1 pencil we have to multiply the cost of 1 pencil with the no. of pencils. For example: in the above table the cost of 1 pencil = Rs 5/-. Therefore the cost of 4 pencils = Rs 5 x 4 = Rs 20/-.

Or, total cost of objects = cost of 1 object x no. of objects.

Similarly, to deduct the cost of 1 object from the cost of some objects we should divide the cost of object by the no. of objects. According to the above table cost of 3 pencils = Rs 15/-. Therefore cost of 1 pencil = Rs 15/3 = Rs 5/-. Therefore,

$$\text{Cost of 1 object} = \frac{\text{total cost of objects}}{\text{No. of objects}}$$

If we know the cost of some objects of same kinds we can deduct the cost of 1 object by multiplying total cost of objects by the no. of objects. It is called unit cost. Similarly, if we know the cost of 1 object we can know the cost of some objects by multiplying the no. of objects with the cost of 1 object. It is called unitary method.

Example 1

If one pen costs Rs 25/-, what will be the total cost of 5 pens?

Here, cost of one pen = Rs 25/-.

No. of pens = 5

Total cost of 5 pens = Rs 25/- x 5/- = Rs 125/-.

Cost of 5 pens = Rs 125/-

Exercise 14.1

1. **Find out the total cost:**

	Unit cost	No. of objects
(a)	Rs 20	15
(b)	Rs 400	45
(c)	Rs 15.50	24
(d)	Rs 250.50	64

2. **Find out the unit cost:**

	No. of objects	Total cost
(a)	10	Rs 250
(b)	32	Rs 672
(c)	60	Rs 5,460
(d)	234	Rs 18,720

3. If the cost of 1 kg of sugar is Rs 40. What will be the cost of 25 kg of sugar?
4. If a business man buys a box of 12 dozen biscuits for Rs. 1860. What will be the cost of 1 dozen biscuits?
5. If Shivaram sold out 45 kg of potatoes at Rs 27. per kg, how many rupees did he get?

6. If the cost of 2 dozen copies is Rs 240, what will be the cost of 1 copy?
7. If 8 people divided and ate up 64 pieces of oranges equally among them how many pieces of oranges did 1 person eat?
8. How many pieces of bread do we need for 15 persons to feed 5 pieces of bread per person?

14.2 Simple Interest

Jairam took a loan of Rs 8000 from Agricultural Development Bank. While returning the loan after a year he paid Rs 8800.

Here, the loan taken from the bank is called principal. So, Rs 8000 is principal. The payment of extra money is called interest. The bank takes interest at certain rate.

Here, Rs 800 has been paid as interest on Rs 8000 for 1 year.

Therefore, annual interest rate $= \frac{800}{8000} \times 100\% = 10\%$

Loan, taken or given for a certain time is called time period. The interest of a principal for a definite time period can be deducted by using unitary method.

Example 1

Rama loaned Rs 100 to Bishnu at the annual interest rate of 10%. Find out how much interest Rama will get after 3 years?

Here, rate interest is 10%

Interest of Rs 100 for 1 year = Rs 10

Interest of Rs 100 for 3 years = Rs 10 x 3 = Rs 30

Rama will get Rs 30 as interest in 3 years.

Example 2

Suraj deposited Rs 400 in the bank. At the annual rate of 5%, how much interest did he get after 1 year?

Here, at the interest rate of 5% per annum,

Interest of Rs 100 for 1 year = Rs 5

Interest of Rs 1 for 1 year = Rs $\frac{5}{100}$

Interest of Rs 400 for 1 year = Rs $= \frac{5}{100} \times 400 = 20$

Suraj got the interest of Rs 20 in 1 year.

Exercise 14.2

1. Find the simple interest on Rs 500 for 1 year at 10% per year.
2. How much is the simple interest on Rs 1200 for 1 year at 10% per year?
3. What will be the interest on Rs 100 for 5 years at 7% per year?
4. What will be the interest on Rs 100 for 7 years at 11% per year?
5. How much is the simple interest on Rs 100 for 5 years at 8% per year?
6. How much is the interest on Rs 600 for 1 year if the interest of Rs 100 is 12% per year?
7. What will be the simple interest on Rs 400 for 7 years at Rs 48 per year?

Lesson 15

BILL AND BUDGET

Bill:

Look at the bill Rupa received from Siddheswar Khadya Bhandar and answer the questions

SIDDESWOR KHADYA BHANDAR				
Dawa, Bhojpur				
Customer's name Rupa Pariyar			Bill #: 0061	
Dawan Mahariya Bhojpur			Date: 0636/07/20	
s.#	Particular	Qty.	Rate (Rs)	Amount
1.	Sugar	5 kg	70.00	350.00
2.	Rice	7 kg	30.00	210.00
3.	Flour	3 kg	40.00	120.00
4.	Dal Musuro	2 kg	80.00	160.00
4.	Gram	1 kg	60.00	60.00
Total				900.00

E 2 O.E.


Sold by: Bimala

- What is the name and address of the customer?
- Who is the seller who gave away the bill?
- Which item did Rupa buy most?
- How much did Rupa pay in total?
- Can you too prepare such a bill?
- How did the total cost of rice become Rs 210?

Budget:

We need budget to operate a family, organization, office etc. We have to estimate or plan on how to spend money in relation to our income. This helps us to do what we need to do. For this we have to plan our budget in the beginning. Look at the example:

The following is the annual budget of Ramsevak Tharu's family. Look at the budget and discuss the questions;

Income		Expenditure	
Source	Amount(Rs)	Heading/Title	Amount(Rs)
Vegetable	7000	Food Items	7000
Chicken	10000	Clothes	4000
Milk	8000	Education	12000
Goats	9000	Miscellaneous	5000
Total	34000		28000

- Which is greater Ramsevak's family's income or expenditure?
- How much does the family save in a year?
- What's the most and least source of their income?
- On what titles do they spend the most and the least?
- Can you also prepare such a budget for your family?

Ramsevak's family has got more income than income. So the family can make a surplus. If our expenditure is more than income we have to manage the shortage of amount from some other sources.

Tips for the teacher: Show real bills (reduction, vat, tax excluded) discuss and ask students to prepare them. Give an example of budget, discuss, ask them to prepare one for their families and give necessary knowledge and skills about budget preparation.

Exercise

- (1) Read the following one day price-list of the wholesale vegetable market of kalimati and answer the following questions:

Price-list	
Item	Cost per kg
Cauliflower	Rs 30
Cabbage	Rs 25
Tomato	Rs 40
Bean	Rs 35
Pea(green)	Rs 32
Bitter gourd	Rs 36
Chilly	Rs 60
Carrot	Rs 24
Radish	Rs 18

- What is the cheapest vegetable?
 - What is the most expensive vegetable?
 - What is the difference in cost between pea(green) and bean?
 - How much should Hari pay the shopkeeper for 2 kg carrot?
- (2) Four persons bought vegetables respectively according to the lists A, B, C and D. How much did each of them pay? Show it by making bills

(a)

2kg cauliflower
1kg potato
2kg radish

(b)

2kg bean
1kg carrot
2kg cabbage

(c)

1kg chilly
3kg radish
1kg Carrot
1kg potato

(d)

2kg bitter gourd
2kg potato
1kg radish
1kg chilly

e. Which of the above bills is the most expensive?

(2) The following is the annual budget of Janata Primary School, Mugu. Look at it and write answers to the following questions.

Income		Expenditure	
Source	Amount Rs	Titles	Amount Rs
District Education	2,00,000		
Salary	2,30,000		
Aid from V.D.C	50,000	Repair/Renovation	25,000
Local Donation	20,000	Stationary	12,000
Others	25,000	Educational Materials	5,000
		Miscellaneous	2,000
Total	2,95,000		2,74,000

- What is the largest source of income of the school?
- On what is largest expenditure?
- Which is greater income or expenditure of the school?
- How much is the annual surplus of the school?

Lesson 16

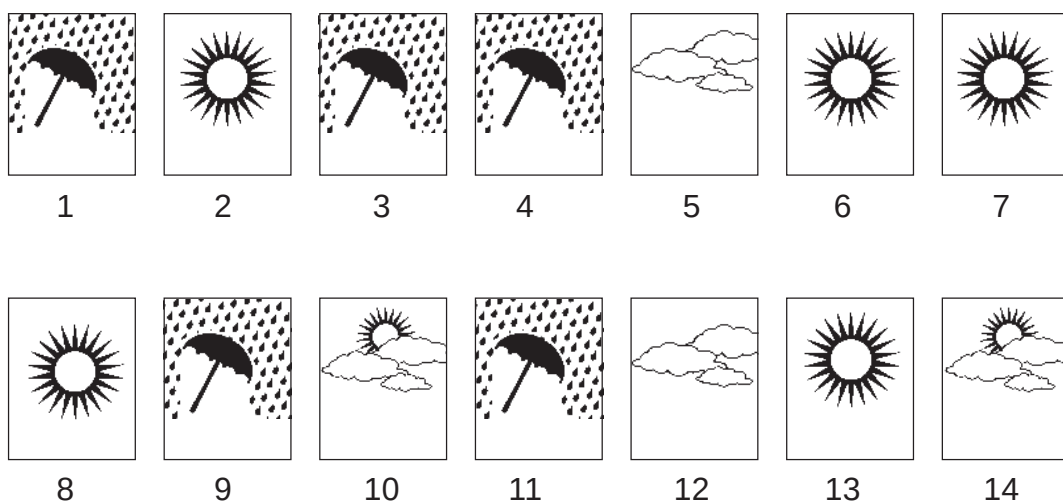
STATISTICS

16.1 Introduction:

Notices or information are expressed on charts and pictures in many places. We can see the food varieties and their cost on the menu of the hotels and restaurants. For the items bought and their cost we receive bills from the shop. We have seen the progress report, annual calendar, budget etc of so many offices, schools and factories in the form of chart or graph. It is easier to understand and reach conclusion of the notices and information expressed in the form of pictures, chart or graph at one close look. For example the following charts shows what the weather was like in the first two weeks of Bhadra. Read the symbols first and then charts:

			
Heavy rain	Cloudy	Sunny	partially cloudy

2 week's weather report of Pokhara (Bhadra 1 - 14).



Now, answer the following questions:

- How many days did it rain in Pokhara in the last 14 days?
- How many days did Pokhara have clear weather in the last 14 days?
- How many days did Pokhara have cloudy weather?
- Which 2 days did Pokhara have partly cloudy weather?
- How many days did it not rain in Pokhara in the last two weeks?
- Which week had more rain in Pokhara?
- How was the weather in Pokhara on the last day of the two weeks?

Exercise

1. Read the price list of the food market in Janakpur and answer the following questions:

Price list	
Food items	Cost(per kg)
Rice short grain	Rs 45
Mansuli rice	Rs 35
Mansoor dal	Rs 90
Gram	Rs 75
Pea(green)	Rs 62
Sugar	Rs 52
Flour	Rs 25
Wheat flour	Rs 20

- What is the cheapest food item?
- What is the dearest food item?
- What is the difference between the cost of green pea & gram?
- How much will Hari pay to the shopkeeper for 2 kg of sugar?

2. In the following table the types and number of vehicles that departed from Sahid Gate between 6 an 7 on Sunday morning are given. Read the table carefully and answer the questions below it.

Types of vehicle	No. of vehicles
Bus	12
Mini Bus	16
Private car	20
Tempo	10
Taxi	22

- Which vehicle was used most?
- Which vehicle was used least?
- If one-third of the buses went outside of the Kathmandu Valley how many buses went outside?
- How many passengers travel by bus if 40 people travel by one bus on average?

Tip to the teacher: Take students to the nearest shop tell them to ask at home and prepare a bill and siscuss on it with them.

16.2.

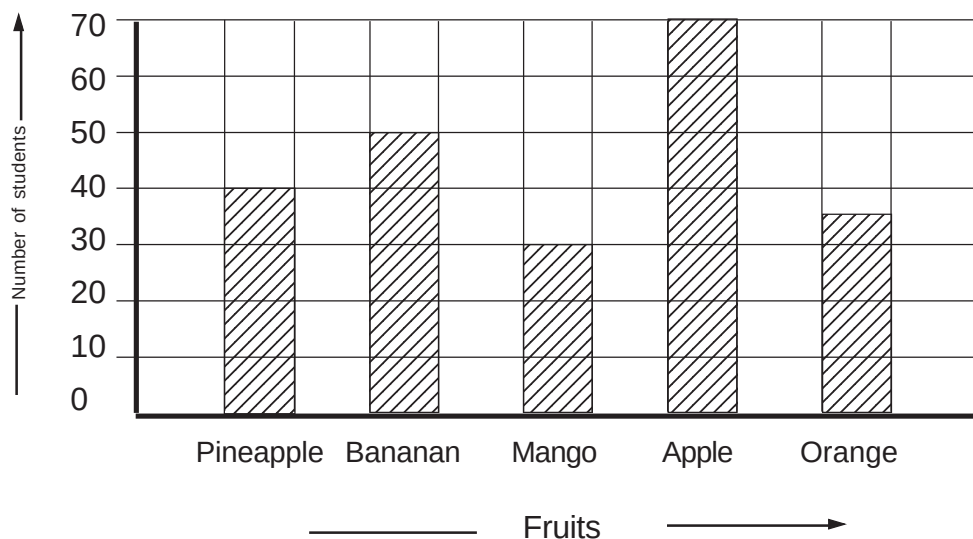
BAR GRAPH

Ramesh was told to collect out information on what types of fruits students like to have during their snack time. Ramesh found out that 40 people liked pineapple, 50 people liked banana, 30 people liked mango, 70 people liked apple and 35 people liked orange. Ramesh thought about how he could express this information so that it is easily understandable by all. He expressed this information on a table.

Fruit	Pineapple	Banana	Mango	Apple	Orange
No. of students who liked	40	50	30	70	35

In this if we put information on a table we can understand things easily.

How can we make facts more understandable, comparable, accessible and attractive? Ramesh thoughts seriously and decided to make a bar graph. On a piece of rectangular paper he went on writing 10,20,30,40 etc on the vertical line and he wrote the names of fruits on the horizontal line. Thus he made the following bar graph:



In the bar graph that Ramesh made the breadth of all the bars are equal. The height of the bar denotes the number of students and the distance between the bars is the same. Thus expression of information or statistics on a bar graph makes it easier to understand and compare many things at a glance.

We should know the following things while making a bar graph:

- We should put variables along the horizontal line.
- We should put numbers on the vertical line.
- The breadth and the distance between two columns should be equal. The numbers on the vertical line should be denoted at equal distance and unit.

Exercise 16.2

Use a graph paper for this:

1. The traffic police collected the following statistics on the types of vehicles that move around Tansen Bazaar between 6 and 9 in the morning:

Name of the Vehicle	Passenger Bus	School Bus	Private Car	Government Vehicles	Taxi
Number	10	7	3	10	25

Draw a bar graph to show types of vehicles on the horizontal line and one unit equal to one vehicle on the vertical line.

2. The following statistics was collected while measuring the height of class 5 students.

Height (cm)	105	106	107	108	109	110
Number	3	12	20	13	7	5

Draw a bar graph showing 1 unit = 1 student on the vertical line.

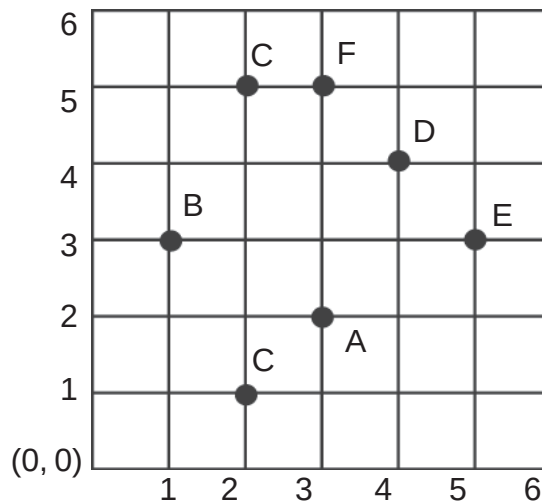
3. The table shows the number of students present at school in a week:

Days	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
No. of students present	25	20	22	18	21	23

Draw a bar graph to represent this information.

16.3 Pair of numbers and coordinates

In graph paper every point can be represented by using a pair of numbers.



Suppose Hari is at point (0,0). If he wants to go to point A then he has to go 3 units right and 2 units above.

We can denote point A by writing (3, 2). In this way the numbers which are used to denote specific points are called pair of numbers or coordinate.

Similarly to go to point E he has to go 5 unit right and 3 units above. The pair of numbers or coordinate of E are (5,3). So in order to go any points from (0,0) then we must have to move right first and then above.

Write down the pair of numbers or coordinate of D.

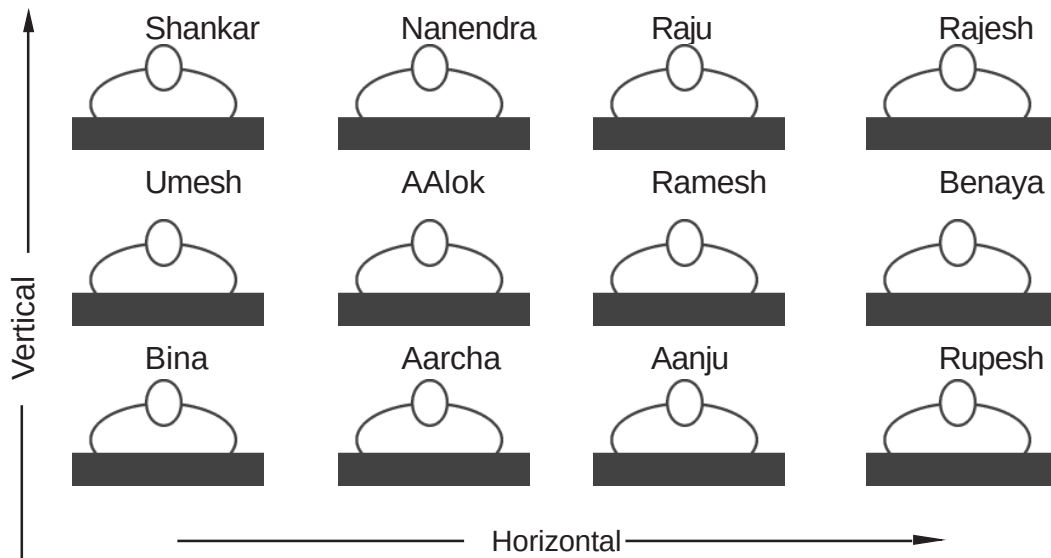
What does the points (1, 3) represents?

Write the alphabetical point from the graph.

Which letter does the point (4, 4) represent?

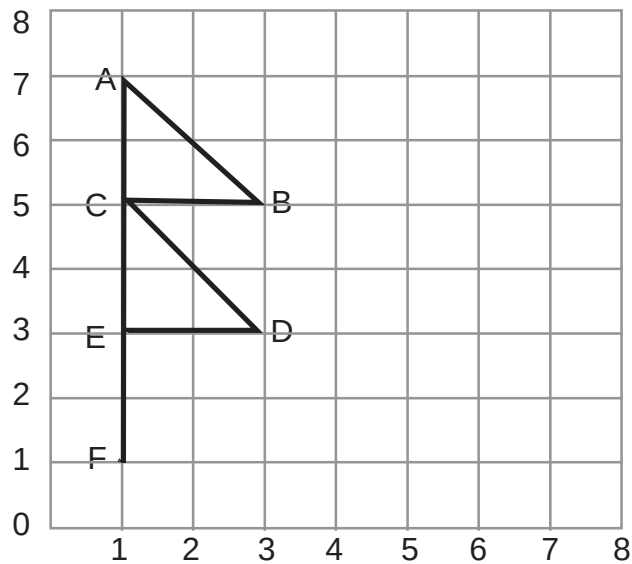
Exercise 16.3

- (1) 12 students of class 5 are arranged to form 3 rows and 4 columns to know the position of any student we have to move left & then up. In this way if we write the coordinate of Ramesh then, first we move 3 students left & 2 students up, so it is found to be (3,2). Similarly, find out the position or points of each of them except Ramesh.

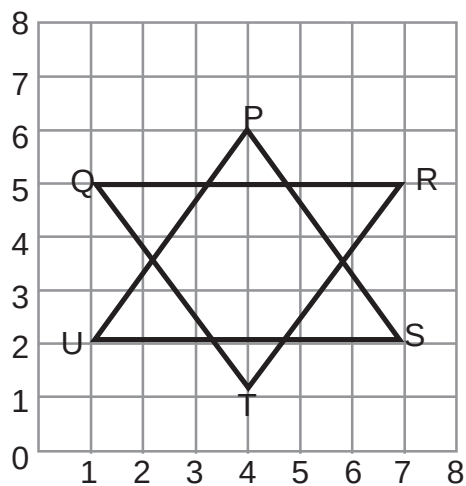


- | | | | |
|------------|-------------|-----------|-----------|
| a. Shankar | b. Narendra | c. Raju | d. Rajesh |
| e. Umesh | f. Alok | g. Ramesh | h. Binay |
| i. Bina | j. Archana | k. Anju | l. Rupesh |

- (2) Look at the picture carefully and find out the pair of numbers of points A, B, C, D, E, F.



- (3) In the given figure write down the points of hexagram.

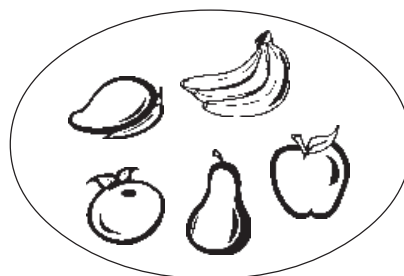


- (4) Draw out the points (2,2), (2,4), (1,4), (3,6), (3,7), (2,7), (4,9), (6,7), (5,7), (5,6), (7,4), (6,4) and (6,2) into a square graph paper and write what does it form after joining all the points.

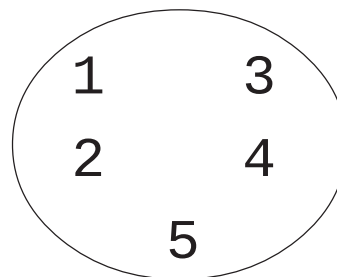
Lesson 17

SET

In the given figure a set of apple, mango, papaya, banana and orange is shown. In this set every members are called elements. Can you write down that how these sets are represented?



There are different ways to represent the sets. The easiest way to represent a set is by writing any English capital letter. for eg: A,B,C,D.....Z. the elements of a set is represented by small letters. We write the set by writing any capital letter first and then giving equals sign and a curly bracket is given { } and the elements of a set is separated by giving comma(.). From the given figure write down the elements of sets by the above mentioned process.



That set is written as {1,2,3,4,5}.let us denote the set by C and these numbers are first five natural numbers. Here,the number of elements in the set is 5and the set is written as:

$C=\{1,2,3,4,5\}$ or $C=\{5,4,3,2,1\}$

In this way the set can be written in any order of the elements present in the set but the repeated elements is not written in a set.for eg:

{S,C,H,O,O,L} is not written but it is written as {S,C,H,O,L}.so an element is written only once.

Example 1

Write down seven days of a week by listing method

$D = \{\text{Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday}\}$

Example 2

What is the set of $E=\{2,4,6,8,10\}$

They are the set of first five even numbers.

Exercise

1. Write down the following sets by listing method using the symbols

- a. Sets of 12 Nepali months.
- b. Sets of five development region of Nepal.
- c. Sets of English vowel letters.
- d. Sets of class 5 books that you should read.
- e. Sets of odd numbers less than ten
- f. Sets of numbers with no remainder when divided to 24.

2. Write down the following sets into sentence

- a. $C = \{\text{cow, buffalo, horse}\}$
- b. $D = \{\text{Kathmandu, Lalitpur, Bhaktapur}\}$
- c. $S = \{\text{Nepal, India, Bangladesh, Srilanka, Pakistan, Afghanistan, Maldieives, Bhutan}\}$
- d. $V = \{a, e, i, o, u\}$
- e. $R = \{I, II, III, IV, V\}$
- f. $E = \{10, 12, 14, 16, 18, 20\}$
- g. $\{5, 10, 15, 20, 25\}$
- h. $\{\triangle, \square, \bigcirc, \square\}$
- i. $F = \{0\}$

ALGEBRA

18.1 Algebraic Expression and its value:

To denote the numbers in algebra we use $"$, Δ , $*$ types of symbols. As in the arithmetic 2×4 represents product of 2 and 4 similarly in algebra also $2X$ represents the product of 2 and X . In algebra $2a, 3b, 5x, 7x$, $"$ etc are called algebraic term. The letters used in algebra are called variables. for eg ,a, b,x and $"$. In the algebraic term $2a$, 2 is the coefficient of a, the value of $2a$ depends upon the value of a.

If $a = 2$ then, $2a = 2 \times 2 = 4$

If $a = 3$ then, $2a = 2 \times 3 = 6$

If $a = 0$ then, $2a = 2 \times 0 = 0$

Note: if there is more than two algebraic terms and are seprated by symbols like (+,—, ÷,×)then such terms are called algebraic expressions.

For example $a + 4, x^2 - xy + y^2, 8a^2bc, \frac{2x+3}{5y}$

Example 1

- Polynomial $" + 4$ means the sum of $"$ and 4.
- Polynomial $x-5$ means the difference of x and 5.
- Polynomial $2a$ means the product of 2 and a.
- $\Delta \div 4$ means the number which comes after dividing Δ by 4.

Example 2

If $x = 4$ and $y = 3$ then find out the value of following terms

(a) $x + 5$ (b) $3x - 2y$ (c) $\frac{3x + 2y}{2y}$

(a) here, $x + 5$

$$= 4 + 5 \quad (\text{putting } x=4)$$

$$= 9$$

(b) here, $3x - 2y$

$$= 3 \times 4 - 2 \times 3 \quad (\text{putting } x = 4 \text{ and } y = 3)$$

$$= 12 - 6$$

$$= 6$$

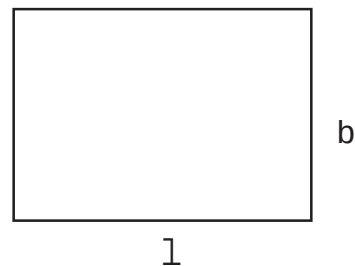
(c) here,

$$\begin{aligned} & \frac{3x + 2y}{2y} \\ &= \frac{3 \times 4 + 2 \times 3}{2 \times 3} \quad (\text{putting } x = 4 \text{ and } y = 3) \\ &= \frac{12 + 6}{6} \\ &= \frac{18}{6} \\ &= 3 \end{aligned}$$

We simplify an algebraic expression by putting the value of variables(s) in it.

Example 3

If a rectangle has its length = l and breadth b , then the perimeter is denoted by algebraic term $2(l+b)$. if we write $P=2(l+b)$ then it is an algebraic formula. Now if $l=5\text{cm}$ and $b=3\text{cm}$ then what is the perimeter?



$$l = 5\text{cm} / b = 3\text{cm}$$

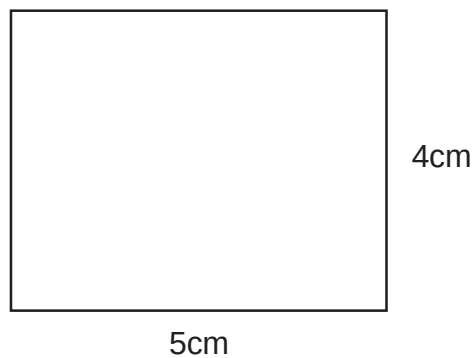
$$\begin{aligned}\text{Here, perimeter } P &= 2(l+b) \\ &= 2(5\text{cm}+3\text{cm}) \\ &= 2 \times 8\text{cm} \\ &= 16\text{ cm}\end{aligned}$$

putting the given
value
 $l = 5, b = 3$



Example 4

If a rectangle has area $A = l \times b$. if $l = 5\text{cm}$ and $b = 4\text{cm}$ then what is the area of rectangle?



$$\begin{aligned}\text{Here area of rectangle } A &= l \times b \\ &= 5\text{cm} \times 4\text{cm} \\ &= 20\text{ cm}^2\end{aligned}$$

$$\text{Hence area } A = 20\text{ cm}^2$$

Note: unit of area is always unit square. If in cm it is in cm square.

Exercise 18.1

(1) Write down the meaning of the following terms:

(a) $\ddot{U} + 3$

(b) $\ddot{U} \div 3$

(c) $\ddot{U} - 5$

(d) $3 \times \ddot{U}$

(e) $3a$

(f) $5m$

(g) ab

(h) $5mn$

(2) Express the following into algebraic term:

a. Sum of a and 3 .

b. Difference of $\ddot{U}$ and 7 .

c. The quotient left after dividing 16 by $\ddot{U}$.

d. Thrice of the product of p and q

e. Thrice the difference of 7 and y

f. Product of twice of a and difference of b

(3) What does the following algebraic terms represent? write in language:

(a) $2(p + q)$

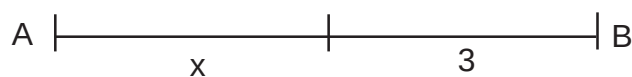
(b) $3(r - s)$

(c) $4(x \div y)$

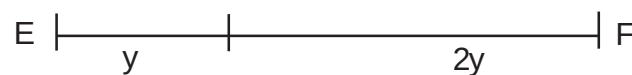
(d) qyz

(4) Write the algebraic expressions for the length of the following lines:

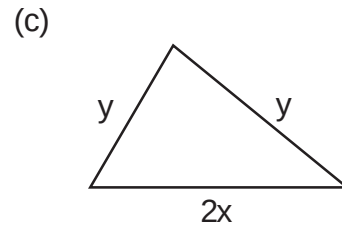
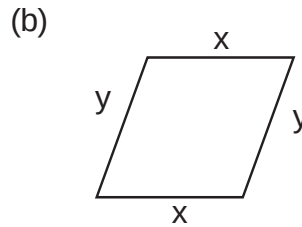
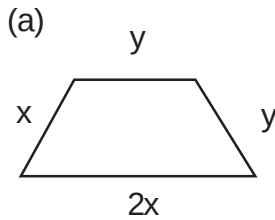
(a)



(b)



(5) Write down the algebraic term to find the perimeter of following figures.



(6) Write down the algebraic expressions for the following phrase:

- Bikash has x apples. He bought 2 more apples. How many apples Bikash has got now?
- If x students out of 500 are absent in a school then find the number of students those are present.
- There were 30 birds in a tree. Other b birds came on the tree. Now how many birds are there in the tree?

(7) What will be the value of the following expressions if we put value $x=2$:

- (a) $3x$ (b) $3 + x$ (c) $5 - x$ (d) $x - 1$
- (e) $3x + 2$ (f) $2x \div 4$ (g) $(5x+2) \div 4$ (h) $\frac{12x + 4x}{2x}$

(8) $a = 1$, $b = 3$ and $c = 5$, then find the values of the following expressions:

- (a) $a + b + c$ (b) $2a + 3b + 4c$ (c) $b + c - 8a$
- (d) abc (e) $10a + 10b + 10c$ (f) $ab + bc + ca$

18.2 Addition & subtraction of algebraic expressions

Example 1

What is the sum of $3a$ and $5a$?

Here, $3a = a + a + a$ (three times a)

And $5a = a + a + a + a + a$ (five times a)

So, $3a + 5a = a + a + a + a + a + a + a + a$ (eight a)
 $= 8a$

In other way,

$$\begin{aligned} 3a + 5a \\ = (3 + 5)a \\ = 8a \end{aligned}$$

If we add those in column then,

$$\begin{array}{r} 3a \\ + 5a \\ \hline \end{array}$$

$8a$, we will get same result.

When we add 5 in 3 it becomes 8. Therefore in $3a$ we add $5a$ it becomes $8a$



Example 2

Subtract $3a$ from $5a$

Here, subtracting $3a$ from $5a$ is to subtract 3 times a from 5 times a

$$5a - 3a = (5-3)a = 2a$$

If we keep those in column then,

$$\begin{array}{r} 5a \\ - 3a \\ \hline \end{array}$$

$2a$, we will get same result.

Form the above examples we see that -

5a and 3a are both common terms. The addition and subtraction of the common terms means the addition and subtraction of the multiples of the variable. Similarly, in the addition or subtraction of the two or more than two algebraic expression we add or subtract the common terms only. Look at the following examples:

Example 3

Simplify:

$$3ab - 4bc + 7ab$$

$$3ab - 4bc + 7ab$$

$$= 3ab + 7ab - 4bc \quad (\text{keeping the common terms together})$$

$$= (3 + 7)ab - 4bc$$

$$= 10ab - 4bc \quad (\text{ab and bc are not common terms so they can't be subtracted, but are only expressed in algebraic terms})$$

Example 4

Add: $2x + 8y - 10z$ to $5x - 7y + 12z$

$$\text{Here, } 2x + 8y - 10z + 5x - 7y + 12z$$

$$= 2x + 5x + 8y - 7y - 10z + 12z$$

$$= (2 + 5)x + (8 - 7)y + (12 - 10)z$$

$$= 7x + y + 2z$$

If we keep those in column and add then-

$$\begin{array}{r} 2x + 8y - 10z \\ + 5x - 7y + 12z \\ \hline 7x + y + 2z \end{array}$$

If we want to add similar type of numbers we have to put them in column. I got it.



Example 5

Subtract $3a + 4b + 7c - (2a + 3b)$

Here, $3a + 4b + 7c - (2a + 3b)$

$$\begin{aligned} &= 3a + 4b + 7c - 2a - 3b \\ &= (3a - 2a) + (4b - 3b) + 7c \\ &= (3 - 2)a + (4 - 3)b + 7c \\ &= a + b + 7c \end{aligned}$$

While subtracting like this if similar type of number has sign + then we have to convert it to - and if - then to +



If we keep those in column and subtract then -

$$\begin{array}{r} 3a + 4b + 7c \\ 2a + 3b \\ \hline a + b + 7c \end{array}$$

Example 6

To make $7a + 2b$ from $5a + 6b$ how much should we add. Think in this way, to make 7 from 5 how much should we add? We can easily say 2. But what mathematical simplification is hidden? We must consider that. Here given number 5 and product number is 7 so to make 7 how much should we add? We know that the number to be added is 2, and while subtracting 5 from 7 we get 2, so by subtracting given number to the product number we get the required number. Therefore in the above problem -

$$\begin{aligned} &7a + 2b - (5a + 6b) \\ &= 7a + 2a - 5a - 6b \\ &= 7a - 5a + 2b - 6b \\ &= 2a - 4b, \text{ is the given number} \end{aligned}$$

Exercise 18.2

(1) Simplify

(a) $a + 3a$

(b) $3m + 4m$

(c) $45p - 13p$

(d) $17n - 3n$

(e) $3x + 4x - 5x$

(f) $5cd - 10cd + 12cd$

(g) $30pr - 35pr + 5pr$

(h) $17x + 3 + 5x - 2$

(e) $a + b + a + c$

(2) Add:

$$\begin{array}{r} (a) \quad 13c \\ + 8c \\ \hline \end{array}$$

$$\begin{array}{r} (d) \quad 10c \\ + 8c \\ \hline \end{array}$$

$$\begin{array}{r} (c) \quad 4a + 5b \\ + 4a + 7b \\ \hline \end{array}$$

$$\begin{array}{r} (d) \quad 4m + 3n \\ + 2m + n \\ \hline \end{array}$$

$$\begin{array}{r} (e) \quad 9ab + 5bc \\ + 7ab - 3bc \\ \hline \end{array}$$

$$\begin{array}{r} (f) \quad 16ab + 14cd \\ + 2ab - 10cd \\ \hline \end{array}$$

(3) Subtract:

(a) $7m$ from $2m$

(b) $16x$ from $9x$

(c) $7x + 9y$ from $3x + 4y$

(d) $3pq - 2qr$ from $2pq - 4qr$

(e) $14ab - 7pc$ from $9ab + 6pc$

(f) $12mn + 10ny$ from $10mn + 2ny$

(4) Add the following expressions by keeping them in columns:

(a) $5x + 7y - z$

and $6x + 3y - z$

(b) $m - 4n + 3$

and $7m + 5n + 2$

(c) $17ab - 13bc + 8ca$

and $13ab + 2bc - 6ca$

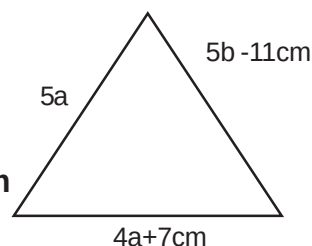
(c) $12x - 16y + 2z$

and $2x + 8y - z$

(5) **Subtract the second term from the first by vertical method:**

- a. $5x + 7y + 12z, 3x + 2y + 10z$
- b. $3a + 5b + 7c, a + 3b + c$
- c. $13bc + 11ab + 4ac, 7bc - 2ab + 3ac$
- d. $3x - 6y + z, x - 4y + 3z$

(6) If $a = 2$ cm and $b = 9$ cm what is the perimeter of the given triangle?



(7) **What must be added to $x + y$ so that the sum may be $7x + 3y$?**

(8) **By how many times is $7x - 4y + 3z$ greater than $3x - 6y + z$?**

(9) **What must be added to $4a + 36$ so that the sum may be $7a + 116$?**

(10) **What must be added to $2a + 7b + 4c$ so that the sum may be $7a + 7b + xc$?**

18.3 Equation

Football team A scored 5 goals in the first half of the football match and the score was equal to team B. Here, we may suppose x as the score of team B and because the goals scored by team B is equal to team A we may write $x = 5$. $x = 5$ is an equation. In the second half both the teams scored 2 goals each and the game was draw. Adding the scores of both halves team B scored $x + 2$ goals and team A scored $5 + 2$ but both the teams scored equal.

Hence, $x + 2 = 5 + 2$

Or $x + 2 = 7$ is called an equation.

Thus we can add the same number to both sides of an equation. This is called coefficient.



Let's think it in a different way. Total game equation is $x+2 = 7$. How can we find out the number of goals x in the first half ?

We know that both team scored 2 goals each in the second half. So, to know the scores in the first half we can subtract 2 from both sides of the equation.

Hence, $x+2-2 = 7-2$

Or, $x = 5$ is the score of the first half. We say that 5 is the solution of the equation $x+2 = 7$. Because team A scored 5 goals and equalled its score with B. Thus we can subtract the same number from both sides of the equation, can't we? Can we also multiply or divide by the same number from both sides of the equation?

Let's see another example:

Shiva had 6 apples. Kailash has got the same number of apples too. Here if x denotes the number of apples with Kailash and because Shiva also has equal number of apples we can write $x=6$. It is an equation. Now, if Shiva & Kailash both ate up half of their apples equal number of apples will remain with them.

Here, $\frac{1}{2}$ of 6 means $6 \times \frac{1}{2}$ and $\frac{1}{2}$ of x means $x \times \frac{1}{2}$. Because both of them have equal number of apples,

$$x \times \frac{1}{2} = 6 \times \frac{1}{2}$$

$$\text{Or } \frac{x}{2} = 3$$

It means $\frac{1}{2}$ of x means 3 which is the number of apples remaining with Shiva & Kailash. Here, multiplication by the same number $\frac{1}{2}$ on both sides of the equation $x = 6$ is the same as the division by the same number 2 on both sides of the equation. It means we can multiply or divide by the same number on both sides of the equation.

On the basis of the discussion above learn the following axioms about equation:

- If the same number is added to both sides of an equation, the two sides are still equal.
If $x = 5$, $x + 2 = 5 + 2$
- If the same number is subtracted from both sides of the equation, the two sides are still equal.
If $x = 5$, $x - 2 = 5 - 2$
- If both sides of an equation are multiplied by the same number the two sides are still equal.
If $x = 6$, $2 \times x = 2 \times 6$
- If both sides of an equation are divided by the same number the two sides are still equal.

$$\text{If } x = 6, \frac{x}{3} = \frac{6}{3}.$$

Example 1

Solve: $X + 5 = 7$

Here, $X + 5 = 7$

Or, $X + 5 - 5 = 7 - 5$ (subtracting 5 from both sides)

Or, $X = 2$.

Example 2

Solve: $Y - 7 = 11$

Here, $Y - 7 = 11$

Or, $Y - 7 + 7 = 11 + 7$ (adding 7 on both sides)

Or, $Y = 18$.

Example 3

Solve: $3x + 2 = 14$

Here, $3x + 2 = 14$

Or, $3x + 2 - 2 = 14 - 2$ (Subtracting 2 from both sides)

Or, $3x = 12$

Or, $\frac{1}{3} \times 3x = \frac{1}{3} \times 12$ (multiplying both sides by $\frac{1}{3}$)

Or, $x = 4$.

To find out whether the equation is correct or not we can check it by putting $x = 4$ in the equation.

Here, $3x + 2 = 14$

Or, $3 \times 4 + 2 = 14$

Or, $12 + 2 = 14$

Or, $14 = 14$ which is true.

In this example $3x = 12$ to find out the value of x multiplication is done by $\frac{1}{3}$ of 3.

3 multiplied by $\frac{1}{3}$, $3 \times \frac{1}{3} = 1$. 3 multiplied by $\frac{1}{3}$ or 3 divided by 3 is the same.

Therefore,

$3x = 12$ divided by 3 on both sides

$$\frac{3x}{3} = \frac{12}{3}$$

$x = 4$, Its meaning and value is the same.

In short,

Subtracting 2 from each side of the equation, $3x + 2 - 2 = 14$

Or, $3x = 12$ dividing both sides by 3

$x = \frac{12}{3}$ which means $x = 4$

$$\therefore 3x + 2 = 14$$

$$\text{Or, } 3x = 14 - 2 = 12$$

$$\text{Or, } x = \frac{12}{3} = 4$$

Therefore,

$$X = 4.$$

Example 4

Simplify and solve the like terms:

$$5x + 4 = 2x + 7$$

$$\text{Here, } 5x + 4 = 2x + 7$$

$$\text{Or, } 5x - 2x + 4 = 2x - 2x + 7 \text{ (subtracting } 2x \text{ from both sides)}$$

$$\text{Or, } 3x + 4 = 7$$

$$3x + 4 - 4 = 7 - 4 \text{ (subtracting 4 from each side)}$$

$$\text{Or, } 3x = 3$$

$$\text{Or, } \frac{3x}{3} = \frac{3}{3} \text{ (dividing both sides by 3)}$$

$$\text{Therefore, } x = 1$$

Exercise 18.3

(1) Find the value of x in each of the following equation:

$$(a) x - 5 = 12$$

$$(b) x + 7 = 10$$

$$(c) x - 6 = 2$$

$$(d) x + 10 = 21$$

$$(e) 9 = x - 4$$

$$(f) 48 = x + 15$$

(2) **Solve and check if it is true or not:**

(a) $2x = 4$

(b) $3y = 9$

(c) $5k = 10$

(d) $\frac{1}{2}m = 6$

(e) $\frac{3}{4}n = 12$

(f) $3x = \frac{1}{3}$

(3) **Simplify and solve:**

(a) $2x + 3x = 15$

(b) $3m + m = 12$

(c) $2y + 5y = 14$

(d) $8x = 24 + 5x$

(e) $5z - 2z = 4$

(f) $6z = 9 + 2z$

(g) $4m - 8 = 2m$

(h) $5p - 3 = 2p$

(4) **Solve:**

(a) $5x + 7 = 22$

(b) $2x - 3 = 9$

(c) $3x + 11 = 14$

(d) $3x - 4 = 18$

(e) $2x + 5 = 9$

(f) $5x + 4 = 2x + 6$

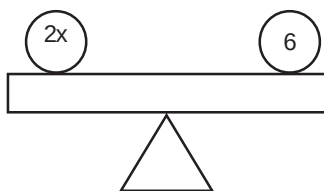
(g) $x - 100 = 0$

(h) $7x + 5 = 9 + 5x$

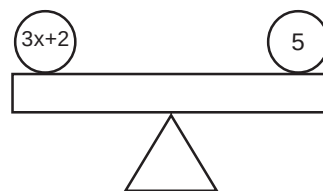
(i) $7x - 2 = 4x + 10$

(5) **If each of the following see-saw are balanced being parallel to the ground, what is the value of x?**

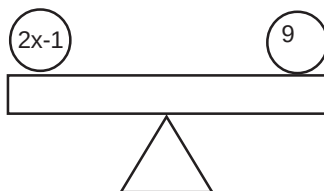
(a)



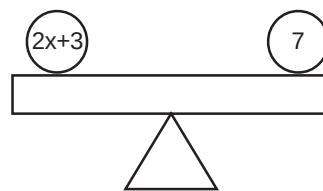
(b)



(c)



(d)



18.4 Use the equation

One of the several uses of equation is to solve problems that come across in our daily life. For this the given problem which should be changed into mathematical language in other words this should be solved by changing it into equation. Here, the unknown terms can be expressed as x and y . See the following example:

Example 1

If 5 is added to a certain number and the result is 18, what will be the number?

Here, if the required number = x so $x + 5 = 18$

Or, $x + 5 - 5 = 18 - 5$

Or, $x = 13$ is the required number.

Example 2

What is the number, which when added to its two times gives 15.

Here, if the required number = x , $x + 2x = 15$

Or, $3x = 15$

Or, $1/3 \times 3x = 1/3 \times 15$

Hence, $x = 5$

Exercise 18.4

Write each of the following open sentence in the form of equation and solve:

- (1) After spending rupees 5 if Sunil has got rupees 15, how many rupees did he have?
- (2) If rupees 5 is added to two times of Ram's money the sum is 17 rupees, how many rupees does he have?

- (3) The number of boys in a school is two times more than that of girls. If there are 300 students in the school, what is the number of girls?
- (4) If $\frac{1}{3}$ of a work takes 20 days to complete, how many days does it take to do the whole work?
- (5) If a certain number and its half is 30, what is the number?
- (6) Pokhara had y ml rainfall on Saturday and the next day it had $(y-1)$ ml rainfall. If the total rainfall was 43 ml, how many ml of rainfall did Pokhara have on Saturday?
- (7) One stick is $2x$ meter and the other is $x + 2$ meter in length. If the total length is 17 meter, what is the length of each stick?